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MILITARY SPECIFICATION

TRUCK, LIFT, FORK, DIESEL, PNEUMATIC-TIRED, GENERAL PURPOSE, 6,000-POUND CAPACITY AT 24-INCH LOAD CENTER

This limited coordination Military Specification has been prepared by the Naval Supply Systems Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-T-40632. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type and size, heavy duty, general purpose, pneumatic-tired, diesel-powered, forklift truck with a capacity of 6,000 pounds at 24-inch load center.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	- Plastic Film, (Polyethylene, Thin Gage).
O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
QQ-L-225	- Leaded Tin Bronze Castings and High Leaded Tin Bronze Castings.

FSC-3930

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QQ-P-416	- Plating, Cadmium (Electrodeposited)
QQ-S-698	- Steel, Sheet and Strip, Low Carbon
QQ-S-700	- Steel, Sheet and Strip, Medium Carbon
QQ-S-741	- Steel, Carbon, Structural Shape, Plate and Bar
QQ-S-781	- Steel Strapping, Flat
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-485	- Enamel, Semi-Gloss, Rust-Inhibiting
TT-E-527	- Enamel, Alkyd, Lusterless
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
UU-T-81	- Tags, Shipping and Stock
VV-F-800	- Fuel Oil, Diesel
VV-H-910	- Hydraulic Fluid, Non-Petroleum Base, Automotive
VV-L-751	- Lubricants; Chain, Exposed-Gear, and Wire-Rope
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Box, Fiberboard
PPP-T-60	- Tape; Pressure-Sensitive Adhesive, Waterproof - for Packaging

Military

MIL-P-116	- Preservation, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed; Flexible
MIL-S-207	- Sulfuric Acid, Electrolyte: Packaging, Packing and Marking for Shipment and Storage of
MIL-P-514	- Plates; Identification, Transportation Data and Blank
MIL-T-704	- Treatment and Painting of Material
MIL-L-2104	- Lubricating Oil, Internal-Cumbustion Engine, Heavy-Duty

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MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose

MIL-L-6085 - Lubricating Oil; Aircraft Instrument, Low Volatility

MIL-L-10295 - Lubricating Oil, Internal-Cumbustion Engine, Sub-Zero

MIL-L-10324 - Lubricating Oil, Gear, Sub-Zero

MIL-G-10924 - Grease, Automotive and Artillery

MIL-E-11276 - Engine, Diesel, Industrial, Medium-and High Speed, General Specification for

MIL-C-11755 - Compound, Antifreeze, Arctic-Type

MIL-H-15424 - Hand Tools, Packaging of

MIL-I-16910 - Interference Measurement, Electro-Magnetic, Methods and Limits

MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative

MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw

MIL-L-45199 - Lubricating Oil, Internal Cumbustion Engine (High Output Diesel)

MIL-L-46002 - Lubricating Oil, Contact & Volatile corrosion Inhibiting

STANDARDS

Federal

FED-STD-595 - Colors

Military

MIL-STD-100 - Engineering Drawing Practices

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U. S. Military Property

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MIL-STD-1261	- Welding Procedures for Constructional Steels
MS-15001	- Fittings, Lubrication, Hydraulic, Surface Check
MS-15002	- Fittings, Lubrication (Hydraulic) Surface Check, Straight Threads, Steel, Type II
MS-15003	- Fitting, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III

DRAWINGS

NAVAL SUPPLY SYSTEMS COMMAND

3486	- Truck, Lift, Fork, Diesel, Pneumatic - Tires, General Purpose, 6,000 Pound Capacity at 24-Inch Load Center

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

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(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

THE TIRE AND RIM ASSOCIATION, INC.

Yearbook of the Tire and Rim Association, Inc.

(Application for copies should be addressed to the Tireand Rim Association, Inc., 2001 First National Tower, Akron, Ohio 44308.)

AMERICAN WELDING SOCIETY

STANDARD QUALIFICATION PROCEDURE OF THE AMERICAN WELDING SOCIETY

(Copies of the Welding Society of American Codes may be obtained upon application to the American Welding Society, 33 West 39th St., New York, N. Y., 10018).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Welding Qualifications of the American Society of Mechanical Engineers

(Copies of ASME Codes may be obtained upon application to the American Society of Mechanical Engineers, 345 E. 47th St., New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Description. The forklift truck (hereinafter called truck) shall be diesel-engine driven and operator selected 2- and 4-wheel steer and 2- and 4-wheel drive. The truck shall have an oscillating frame and a hydraulic-powered forklift mechanism with a capacity of lifting 6,000 pounds at a 24-inch load center (rated load) to a height of 144 inches. Load center is defined as the horizontal distance from the front face of the forks to the center of gravity of the load. The truck shall be capable of traversing rough terrain, fording water to a depth of 5 feet without kit, and being transported by air and rail. The truck shall perform all operations specified herein without failure, permanent deformation or malfunction to any component or structural member when tested as specified in section 4. The truck shall be in accordance with Naval Supply Systems Command

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Drawing Series 3486 and as specified herein. Where there is a conflict between the operational or environmental conditions specified on the drawings and those specified herein, the specification requirements will take precedence.

3.1.1 Alternate components. Components cited herein and on the referenced drawings are mandatory. Components offered in lieu of these specified hereinafter and on the referenced drawings as a specific manufacturer's part number shall be physically interchangeable without modification or change to any other component, functionally equal, of equal or better quality, with the manufacturer's part number identified. The bidder shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirement specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of preproduction sample manufacture and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.1.2 Drawings. The drawings forming a part of this specification are engineering design drawings. The supplier is responsible for preparing his own shop drawings and manufacturing processes. Where no tolerances or finish marks are indicated on the drawings, the supplier is responsible for establishing tolerances and finishes which will result in a truck meeting the requirements of this specification. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operating of the truck. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. The supplier shall make available to the Government, a complete set of shop drawings with manufacturing processes for inspection and comparison purposes.

3.2 Preproduction models. The supplier shall furnish a completely assembled truck within the time specified in the contract or purchase order to prove, prior to starting production, that his engineering and production methods and his use of alternate components, will produce trucks that comply with the requirements of this specification. Examination and tests shall be those specified herein. The purpose of testing the preproduction sample is to determine whether the item manufactured by the contractor has the performance characteristics required by the specification and that the quality of workmanship has not effected the performance criteria established by the design requirements. Any changes or deviations from the approved preproduction model shall be subject to the approval of the contracting officer. When a change or modification to the drawings listed in Section 2 has been allowed by the contracting officer, the contractor will

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furnish revised reproducible drawings with tolerances and finishes in accordance with MIL-STD-100, to show the changes or modifications allowed. The drawings shall be furnished to the Naval Supply Systems Command, Code 0443, Washington, D.C. 20360. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish trucks conforming to this specification. Subsequent to any preproduction model testing the supplier shall install Government-furnished seals to adjustment devices in such places as directed by the Government representative. These seals shall remain in place during all testing and shall not be removed or altered except by the Government representative. The seals will be removed or replaced to permit maintenance, evaluation, or part replacement when authorized by the Government representative. Unauthorized removal or alteration of the seals at any time during testing shall require rerunning of all tests. The suppliers shall provide the tools and labor necessary for the installation of the seals to the preproduction model.

3.2.1 Material identification. Unless otherwise specified (see 6.2) the supplier shall furnish to the Government, in the form of sketches or outline drawings (dimensions not required), identification of high strength constructional steels (steels requiring special process controls as defined in MIL-STD-1261) used in the fabrication of the truck and the recommended welding repair procedures for accomplishing repair of such steels by the Military in technical manuals (see 6.4). Identification shall include type of steel with ASTM designation and grade and location by part or assembly.

3.3 Initial production. The supplier shall furnish a truck for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall meet the requirements of this specification. Unless otherwise specified her in, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the truck by the Government.

3.4.1 Steel.

3.4.1.1 Steel, sheet and strip. Steel sheet and strip, hot or cold rolled, shall conform to FS1009-1020 of QQ-S-698 and FS 1025 of QQ-S-700 unless otherwise specified on the applicable drawings.

3.4.1.2 Steel structural shapes. Structural steel, shall conform to type II, class 1 of QQ-S-741 unless otherwise specified on the applicable drawings.

3.4.2 Fastening devices. Bolts, nuts, washers, cotter pins in the types and sizes specified on the drawing shall be steel and finished as specified in 3.14.2.

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3.4.2.1 Screw threads. Screw threads shall be in accordance with Handbook H-28. Except where otherwise specified herein all hexhead cap screws shall be SAE grade 5 or 8 and marked. Bolts used for fabrication or assembly of major subassemblies that are not normally SAE 5 or 8 in production need not be SAE 5 or 8.

3.5 Design. The fully assembled truck shall be capable of meeting all performance characteristics specified in 3.8 and 3.9.

3.6 Environmental Requirements.

3.6.1 Operating temperature. The engine shall start without preheating within 5 minutes, and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from minus 25° F. to plus 120° F. (See 4.6.11 and 4.6.12).

3.6.2 Storage temperature. The truck shall not be damaged by storage under the following conditions:

Air temperature	Time period
Plus 155° F.	4 hours daily
Minus 65° F.	12 hours (no solar radiation)

3.7 Fuels, lubricants and coolants. The truck shall be tested and furnished with standard Military service products of the type and grade specified in Table I for the applicable ambient air temperature. The truck shall operate as specified herein when serviced with these products.

TABLE I - Service Products

Item Application	Specification	Type & Grade
1. Engine oils		
a. Heavy duty	MIL-L-2104	10, 30
b. Operational and preservative (preservation only)	MIL-L-21260	Grade 1, 2, & 3
c. Arctic	MIL-L-10295	One grade
2. Fuel		
Diesel	VV-F-800	DF-1, DF-2, & DF-A
3. Grease		
Automotive	MIL-G-10924	One grade

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4. Hydraulic-system oil and transmission fluid		
a. Oil	MIL-L-2104	10
b. Arctic (for consistent operation below - 10°F)	MIL-L-10295	one grade
5. Lubricating oils		
a. Chain, gear, wire rope	VV-L-751	A, B, C
b. Universal	MIL-L-2105	75, 80, 90
c. Arctic	MIL-L-10324	One grade
d. Aircraft instrument	MIL-L-6085	One grade
6. Anti-Freeze		
a. Ethylene glycol	O-A-548	Type I
b. Arctic	MIL-C-11755	One grade
7. Brake fluid	VV-H-910	One grade

3.7.1 Lubrication fittings. Lubrication fittings shall conform to MS15001, MS15002, and MS15003. Fittings where not shown on the drawings shall be located in a protected position and shall be accessible to a hand-operated grease gun. Accessibility to fittings shall be provided without the removal of major components.

3.7.2 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.7.3 Lubrication service. The truck shall be tagged on the steering column to indicate which lubricant has been used in each lubricated component.

3.7.3.1 Grease lubrication. All grease lubrication shall be automotive in accordance with Table I, item 3.

3.7.3.2 Oil lubrication. The truck shall be delivered with type or grades of oils for the temperature ranges as specified (see 6.2.).

3.8 Lifting attachments. The truck shall be provided with four lifting attachments positioned to lift the truck in its normal traveling position. The attachments shall be located so that a minimum clearance of 1 inch is maintained between the slings and all exterior parts of the truck. Spreader bars are permitted. Each attachment shall carry not less than 2-1/2 times the normal load imposed on the attachment. The slings shall converge not more than 19 feet above the lowest extremity of the truck. The inside diameter of the eye shall be not less than 3 inches. Structural members to which the attachments are fastened and the fastenings shall withstand all stresses imposed by the attachment without permanent deformation and breakage (see 4.6.13). Where required, structural members

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shall be reinforced to meet the requirements specified herein.

3.9 Performance and dimensional characteristics.

3.9.1 The completely fabricated truck shall meet the dimensional and performance characteristics listed in Table II:

TABLE II. Dimensional and performance characteristics

Characteristic	Requirements
Height, inches maximum (without overhead guard)	92
Length, inches maximum (with fork in retract position)	218
Width, inches (overall), nominal	87
Track, inches (frong and rear), minimum	70
Fork length, inches nominal	48
Ground clearance, inches, minimum	14
Lifting height, inches, minimum	144
Lifting speed, full load, feet per minute, minimum	40
Load capacity, pounds @ 24 inch load center, minimum	6,000
Speed, miles per hour, empty, forward and reverse, minimum	25
Speed, miles per hour, loaded, forward and reverse, minimum	15
Fork tilt, degrees	
Backward, minimum	30
Forkward, minimum	45
Oscillate, degrees, total, minimum	20

3.9.2 Fording conditions. All components, assemblies and systems of the truck which would be damaged or could cause malfunction of the truck when exposed to salt water shall be inclosed in a waterproof housing or jacket. Equipment inclosed in a waterproof housing that must be air vented shall have a vent tube or pipe that extends above the 5 foot fording depth to preclude the entry of water. The truck shall be capable of being started at full water depth. All exposed components subject to contamination or corrosion from fresh or salt water shall be protected by use of a corrosion-resisting plating or a corrosion-resisting treatment, except the wearing surfaces of functional chains, rollers and slides which shall be lubricated in accordance with the technical manual. The application of a corrosion-resisting plating or treatment shall give protection to any part for its functional life. There shall be no evidence of water damage to or water leaking into waterproof housings, waterproof jackets and closed tanks of the engine, transmission, torque converter, fuel system, brakes and hydraulic system or any evidence of water damage or corrosion of any truck part when tested as specified in 4.6.22.2.

3.9.3 Truck stability. The completely assembled truck shall incorporate adequate longitudinal and lateral stability within the following limitations, when tested as specified in 4.6.9.2

a. With no load - Heel of forks 21" front of tires, 24" above ground

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rear axle - 57 %, minimum of the curb weight (see 3.13).

b. With rated load - Heel of forks 21" front of tires, 24 " above ground, rear axle - 21 %, minimum of the gross vehicle weight (GVW).

c. Under (a) and (b) above, the difference in weight between the left and right wheel of either axle shall not exceed 6 % of the total weight on that axle. The position of the forks described in (a) and (b) above is defined as the carry position.

3.9.4 Slope operation.

3.9.4.1 Longitudinal slope. The truck with rated load shall be capable of negotiating a 45% longitudinal slope (see 4.6.9).

3.9.4.2 Lateral slope (static). With the lateral axis of the truck parallel to the truck axles, the truck with rated load shall maintain 4-wheel ground contact on a 30 percent side slope when tested in accordance with 4.6.9.

3.9.4.3 Lateral slope (dynamic). The truck shall be capable of operating on a 30% lateral slope without load when tested as specified in 4.6.9.

3.9.5 Endurance. The completely assembled truck shall be capable of completing the 100 hour load placement test specified in 4.6.22.1(a) and the 300 hour pothole and obstacle course specified in 4.6.22.1(b) without malfunction, or permanent deformation.

3.9.6 Speed. The truck in two wheel drive, front wheel steer with no load shall be capable of operating at a speed of 25 miles per hour (mph) when tested as specified in 4.6.7. The truck in two wheel drive, front wheel steer with rated load shall be capable of operating at a speed of 15 mph in both forward and reverse gears when tested as specified in 4.6.7.

3.9.7 Acceleration. The truck with no load shall be capable of traversing a 100-foot course in 19.0 seconds in low gear and 8.9 seconds in second gear and 7.5 seconds in high gear in both forward and reverse gears when tested as specified in 4.6.8.

3.10 Details of components. All components shall be as specified herein, as shown on the drawings listed in Section 2, and as required by 3.1.1.

3.10.1 Engine. The engine when not Government furnished (3.16.1), shall conform to the requirements specified herein and in MIL-E-11276 and as required by the drawings listed in Section 2.

a. Electromagnetic interference limits shall be in accordance with 3.11.

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b. A model of the basic engine shall have successfully passed the qualification test requirements of MIL-E-11276. The contractor shall have available a certificate of compliance from the engine supplier indicating that the engines furnished comply with the above.

c. Maintenance parts and tools shall be as specified (see 6.2).

d. Spare parts support as required by 3.19.

e. The engine shall be provided with a locking type hand throttle on the instrument panel.

f. Governor - Provision shall be made for sealing the governor. The governor shall be set so that the engine shall develop no more than the maximum output under no load as established by the engine manufacturer.

g. The engine data plate shall include, but not be limited to the following:

1. Manufacturer's specification number.
2. Manufacturer's model number.
3. Manufacturer's engine serial number.
4. Date of manufacture.
5. Bore, stroke, displacement.
6. Firing order.
7. Governor setting under no load....RPM.
8. Maintain oil pressure....lbs.
9. Net continuous BHP....under full load at....RPM.
10. Manufacturer's maintenance manual number.
11. Military specification number.
12. Manufacturer's name.

h. Engine shall be waterproofed to permit operation under fording conditions specified in 3.9.2.

3.10.2 Frame and body assembly. The frame and body assembly shall be in accordance with the applicable drawings with no structural or material deviations permitted unless specifically authorized. Material shall be as specified on the drawings and in 3.4. The frame and body assembly shall be free to pivot on the center suspension pins through the action of a hydraulic powered cylinder as shown on the applicable drawings. Rotational motion of the frame and body assembly about the pivot point shall be not less than 10 degrees clockwise and 10 degrees counterclockwise relative to a longitudinal vertical plane through the center of the truck. The hydraulic cylinder, hose lines and hydraulic controls shall be connected into the hydraulic power systems used for operating the forklift mechanism

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and shall withstand the requirements specified therein. There shall be no evidence of permenent deformation or structural failure of the frame and body assembly during any of the test specified in 4.6.22.1

3.10.2.1 Fuel tank. The fuel tanks shall be as specified on the applicable drawings referenced in Section 2 and capable of passing the test specified in 4.6.4.1.

3.10.3 Exhaust system. The engine muffler and piping shall comply with requirements of applicable drawing. When specified (see 6.2), a spark arresting muffler shall be supplied.

3.10.4 Electrical power system. The electrical systems shall be 24-volt consisting of two batteries, starting motor, generator, regulator, and as specified on the applicable wiring diagram(s) and the drawings listed in Section 2. All wires and cables shall be adequately supported to minimize chafing and provide adequate clearance from moving parts, hot engine parts, exhaust and fuel systems. Cables and wires shall not be supported on oil or grease retaining surfaces or be exposed to drippage of fuel, oil or grease. When exposed to unusual vibration, cables and wiring shall be protected by flexible tubing or suitably harnessed. Grommets shall be provided whenever wiring passes through bulkheads, partitions, or structural members, and shall be suitably fastened to prevent chafing or abrasion. An interlock shall be placed in the starting circuit to prevent starting the vehicle while the transmission is in gear.

3.10.5 Power train. The truck power train shall consist of the torque convertor, transmission and axle assemblies. Each gear, shaft and axle shall be of hardened steel and shall be mounted on anti-friction bearings. All gears shall operate in heavy lubricant (see 3.7) and a standard pressure-grease lubricant system shall be provided for all friction parts not lubricated from the normal transmission. All drive shafts shall be provided with 7C mechanic tapped flanges. The entire power train shall be adequately waterproofed by use of seals to preclude the entry of water as required in 3.9.2. The Government reserves the right to require submission of complete drawings and stress-analysis calculations, including design load assumptions through the complete power train assembly from engine to the wheels at the time of testing the preproduction sample where these components are specifically developed for this specification or have no previous military or commercial usage data. All components of the power train when mounted on the vehicle shall be aligned true one to the other with no binding of any of the components within the train.

3.10.5.1 Torque convertor and transmission. The torque convertor and transmission shall be in accordance with the drawing listed in

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Section 2 and as specified herein. The torque convertor shall withstand up to 30 seconds of wide-open-throttle stall operation without exceeding the fluid temperature limit of 250°F. from an initial temperature of not less than 180°F. (see 4.6.10.1). The transmission shall provide not less than three forward and three reverse gear ratios. All transmission gears shall be in constant mesh. The transmission shall provide for shifting under full engine power through all successive forward and reverse transmission gear ratios. The transmission shall provide controlled application of the clutches to provide smooth shifting without slippage which will damage the clutches. Fluid pressure for actuation of the transmission clutches, for torque convertor supply, and for transmission lubrication shall be supplied by a pump mechanically driven by the engine. Belt and chain drives are not permitted. Fluid shall be filtered by a full-flow filter with a replaceable element having a 25-micron nominal filter rating. All filler pipes shall terminate with a cap. A heat exchanger shall be incorporated in the engine cooling system. Heat exchanger pressure drop under maximum flow conditions shall not exceed 10 psig. The transmission breather shall be located vertically above the five foot water level and at the same level as the engine breather.

3.10.5.1.1 Transmission controls. The transmission control levers shall be mounted on the right side of the steering column. The shift pattern shall be arranged to minimize the possibility of accidental shifting into a reverse gear when shifting between successive forward gears. The shifting lever shall pass through the neutral position when shifting from forward to reverse gear. Means shall be provided within the shift mechanism to allow starter operation only when the shift lever is in neutral position. The 2-to 4-wheel drive selector lever shall be located as shown on the applicable drawing listed in Section 2.

3.10.5.1.2 Rear axle disconnect. The truck shall be equipped with a rear axle disconnect suitable for constant use and which shall be actuated by the operator by hand while seated without requiring force in excess of 20 pounds on the lever.

3.10.5.2 Universal joints and drive shafts. Universal joints and drive shafts between the engine and transmission assembly and the outputs from the transmission to the front and the rear axles shall be in accordance with drawings in Section 2 and shall be of identical make and type. The universal joint operating angles shall be in accordance with the recommendations of the drive line component manufacturer as well as the universal joint manufacturer. The load on the projected area of any journal trunnion shall not be in excess of 10,000 psi. The natural-frequency speed of each drive shaft, including shaft mountings, shall be not less than 120 percent of the maximum shaft speed when the truck is traveling at 120 percent of maximum governed road speed.

3.10.5.3 Front and rear axles. The axles shall be in accordance with drawing listed in Section 2 and be of the steering and driving type,

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front and rear. The axle manufacturer's recommended ratings for torque input, gearing, and beam loading shall be in excess of the actual maximum loads. The axle ratio shall be matched to the other drive-train components to meet or exceed the minimum performance specified in 3.9. Both axles shall be of the multiple-reduction type having the final reduction in the hub ends. Axle components shall be interchangeable with respect to at least the differentials, hub assemblies, bearings, seals, and tie rods. Each differential axle carrier shall incorporate a differential action lock of an automatic positive-binding type. Neither wheel of the axle shall spin-out due to loss of traction. Both axles shall be equipped with steering pivots having adjustable bearings. Steering drive joints shall be of the single-carbon type or the constant-velocity, double-carbon or ball type. The axles shall have a cramping angle of not less than 26 degrees, either side of straight ahead, for the wheel on the inside of the truck's turning circle. The tie-rod ends shall permit fine toe-in adjustment. The steering geometry of caster, camber, and toe-in shall be for optimum handling characteristics and tire longevity.

3.10.5.4 Rims and tires. The truck shall be mounted on four single tires, rims and wheels as shown on the applicable drawing. Installed front and rear tire assemblies shall be demountable and interchangeable. The valve stems shall be accessible for servicing by a tire inflator. Tires shall include commercial antiozonant compounds in the rubber mixture. All tires shall be of the same size and ply rating. Tire pressures shall be in accordance with the tire manufacturer's recommendations for the loads carried on each axle. The front tire pressure shall be determined at GVW and the rear axle tire pressure at curb weight. The tires when inflated to 20 psig maximum front and rear shall show no evidence of failure, deterioration or separations from the wheels when truck is operated over soft terrain as specified in 4.6.10.3 and 4.6.22.2.

3.10.6 Steering systems. The steering system shall be as shown on the applicable drawing and shall provide positive hydraulic power steering control of the truck at all speeds and during all operations. The systems shall provide both cramp steering (rear wheels turning in the opposite direction of the front wheels) and crab steering (rear wheels turning in the same direction of the front wheels) (see fig. 1). As viewed from the operator's normal seated position, when the steering wheel is turned clockwise the front wheels shall turn to the right and when the steering wheel is turned counterclockwise the front wheels shall turn to the left. Steering linkage shall be of the modified Ackermann type with performance priority given to the front wheel and cramp-steering operation. The steering system shall be provided with an adjustable, mechanical, follow-up mechanism to maintain a positive ratio between the amount the steering wheel is turned and the amount the front wheels turn. The control of the rear wheels shall be either hydro-mechanical or hydraulic. The power steering actuation device shall be located so as to absorb all road shock without

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transmitting the shock effect to the steering wheel. The steering cylinders shall be mounted on the axles, or the cradles, and all mechanical shock loads and steering forces shall be absorbed by the members.

3.10.6.1 Steering control system. The hydraulic steering control shall be provided with a servo type steering valve in combination with a hydraulic selector valve to control two double acting power steering cylinders; one for each axle. Steering control for the front wheels shall be transmitted by a combination of mechanical linkage and hydraulic effort. The rear wheels shall be steered by full hydraulic means. Wheels of a stationary unloaded truck shall turn from the extreme right position to the extreme left position or vice versa in not more than 6-1/2 turns of the steering wheel and the turning force required to turn the steering wheel shall not exceed 20 pounds when truck is tested as specified in 4.6.14.3. No damage to the steering wheel or steering components shall result when the truck is tested as specified in 4.6.14.1.

3.10.6.2 Four-wheel steering. When the steering selector lever is positioned for four-wheel steering, the turning radius of the truck about the center of turn, measured from the outside track of the tires, shall not exceed 15 feet when tested as specified in 4.6.14.2. When the desired turning radius has been obtained, axle stops shall be permanently set. Screw or simple shim adjustable devices are not acceptable.

3.10.63. Crabbing. With the steering selector lever set for crabbing, the truck shall crab not less than 20 degrees to the right or left of the longitudinal axis of the truck when tested as specified in 4.6.14.4.

3.10.7 Brakes.

3.10.7.1 Service brake. The service brake shall be air assisted hydraulically actuated, spot-type disk brakes as shown on the applicable drawing and shall use the fluid required by item 7, Table I. The air compression system shall contain an alcohol injection system to insure cold weather operation. The brakes shall be capable of satisfactory operation when completely immersed in water and shall be readily accessible for adjustment and servicing. The master cylinder shall be located in an easily accessible place for servicing and the system shall be designed so that bleeding the lines may be accomplished without removing the wheels. Locking wires and lock washers shall be used on all bolts and studs to prevent loosening due to vibration or any other cause. A brake holding device, manually controlled for locking and releasing shall be installed to provide positive locking of the service brake. Component parts of the brakes shall be provided with protective finish, recommended by the manufacturer, to prevent contamination and corrosion resulting from fording operations in both salt and fresh water. The protective finish shall be capable of protecting all parts for their functional service life.

3.10.7.1.1 Service brake effort. The service brakes shall have sufficient

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frictional area and pressure differential to stop truck with rated load as follows:

(a) Stop truck within a distance of 10 feet when driven rearward down a 45 percent slope at 5 m.p.h when tested as specified in 4.6.15.1.

(b) Stop truck within a distance of 5 feet operating forward on a level surface at 5 m.p.h when tested as specified in 4.6.15.2.

(c) Restrain truck on a 45 percent slope immediately after brakes have been entirely immersed in water when tested as specified in 4.6.9.1 and 4.6.15.3.

3.10.7.1.2 Service brake deceleration. When truck is driven forward at maximum speed while carrying rated load on forks at 30 degree rearward tilt, the deceleration rate shall not exceed a rate which would overturn truck or drop the load when a force of 125 pounds is applied to the brake pedal when tested as specified in 4.6.15.4.

3.10.7.1.3 Service brake holding device. The service brake holding device shall be capable of retaining truck with rated load in the carrying position in a stopped position on a 45 percent slope after full disengagement of the brake pedal when tested as specified in 4.6.9.1 and 4.6.15.3.

3.10.7.2 Parking brake. The parking brake as shown in the applicable drawing shall be an independent mechanical friction brake mounted on the drive shaft and connected by a cable and conduit assembly to a hand lever equipped with a locking device. The hand lever shall be located in the operator's compartment for convenient right-hand operation as shown on the applicable drawing. The parking brake shall be capable of stopping forward and rearward motion of the truck with rated load in carrying position when coasting at 2 m.p.h with the transmission placed in neutral position when tested as specified in 4.6.15.5.

3.10.7.3 Air hose and air control valve. The contractor shall furnish with each truck a 20 foot length rubber or synthetic air hose of 1/4 inch ID. The hose shall be provided with male brass couplings having 1/2 by 20 right hand National Fine threads with a class 3 fit and an air inflation chuck. The air hose shall be coiled and placed in the accessory box located in the left front fender. An air supply valve with chain cap identified as Part No. 221351 manufactured by Bendix Westinghouse (or equal) shall be provided by the contractor and shall be located in the air supply system to provide a tire inflation capability.

3.10.8 Hydraulic powered forklift mechanism. The truck hydraulic powered forklift mechanism shall consist of a telescopic lift arm assembly, fork carriage, fork and rack assembly and hydraulic power system with remote controls conforming to the requirements specified herein and on applicable drawings listed in Section 2, and the mechanism shall be capable of performing the following functions through the action of hydraulic

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powered pistons:

(a) Extend heel of forks forward not less than 21 inches from extreme retracted position when the bot om of the forks are 24 inches above the ground and retract fork carriage onto the frame in transport position.

(b) Lift rated load on forks from the ground to height of 144 inches.

(c) Lift rated load on forks not less than 40 feet per minutes

(d) Tilt rated load on forks not less than 45 degrees forward and 30 degrees rearward.

(e) Displace rated load on forks from one extreme edge of truck to the opposite edge of truck without jerking or excessive vibration.

There shall be no evidence of permanent deformation or failure to any component of the lift mechanism, truck frame and body when tested as specified in 4.6.5 and 4.6.16. Also, there shall be no evidence of breakage, no overflow from the hydraulic tank, or leaks in the hydraulic lines of system components. The oil for the system shall be as specified in Table I; item 4a or 4b as applicable. All component materials shall be compatible with these oils and shall operate as specified herein throughout an oil temperature range of minus 25°F. to plus 225°F. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100°F. above ambient temperature during all operations specified herein. The hydraulic power system shall be capable of holding forks with rated load in a maximum lift position for a period of 5 minutes without downward drift of forks exceeding 1 inch or longitudinal tilt position changing more than 1/2 degrees and rotation of frame and body changing more than 1/4 degrees when tested as specified in 4.6.16.3(c). All piston rods shall be provided with a hard chrome plate finish. Surfaces shall be thoroughly cleaned prior to plating in accordance with normal commercial practice. Plating shall be electro-deposited hard chromium finish, having uniform thickness of not less than 0.0005 of an inch. The entire hydraulic system consisting of pumps, lines, valves, cylinders, tanks shall, after assembly, be free of all metal particles, dirt, rust and foreign material that would contaminate the hydraulic fluid or score the hydraulic system components including the pump. The hydraulic pump filter shall be capable of passing the hydraulic fluid at a rate not less than the maximum delivered by the hydraulic pump. All hydraulic components shall be inspected for cleanliness prior to assembly of the components. The presence of foreign material such as metal chips, grindings, dirt, rust or sand shall be cause for rejection of the components.

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3.10.8.1 Fork carriage, fork and rack assembly. The fork carriage, fork and rack assembly shall conform to the requirements specified in the applicable drawing. The forks shall be 48 inches long. The side shifting function shall shift forks simultaneously by hydraulic power. The removable fork rack shall be capable of withstanding a thrust load equal to rated load without permanent deformation or failure when tested as specified in 4.6.17.

3.10.8.2 Pressure-relief valve. The pressure-relief valve in the hydraulic power system shall be as shown on the drawing listed in Section 2.

3.10.8.3 Hydraulic accumulator. A hydraulic accumulator as shown on the drawing listed in Section 2 shall be provided in the hydraulic power system to reduce lift mechanism shock when lowering valve lever is closed abruptly or traveling over rough terrain when tested as specified in 4.6.16.3(c) and 4.6.22.1.

3.10.8.4 Hydraulic hose lines. The hydraulic hose lines shall be in accordance with drawing listed in Section 2 and shall have sufficient flexure to allow forklift mechanism to perform all maneuvers specified in 3.10.8. All hydraulic hoses on the exterior of the forklift mechanism shall be properly protected to prevent damage from abrasion, kinking, pinching and any other damage during the working cycle.

3.10.8.5 Hydraulic controls. The truck hydraulic powered system shall be provided with actuating control levers located next to the operator's compartment for easy right-hand operation. The arrangement of control levers shall be as specified on the applicable drawing and a spacing of 3/4 inch minimum clearance shall be maintained between levers at all times. The control levers shall operate in a forward and rearward motion and shall be self-centering. All control knobs shall be spherical in shape and the fork actuation shall be as indicated in Table III.

TABLE III. Control Knob and Fork Actuation

Control	Motion	Fork Actuation
Lift	Rearward	Raise the fork and carriage assembly.
	Forward	Lower the fork and carriage assembly.
Tilt	Rearward	Tilt forks and carriage assembly rearward.
	Forward	Tilt forks and carriage assembly forward.
Reach	Rearward	Retract forks and carriage assembly.
	Forward	Extend forks and carriage assembly.
Lateral displacement	Rearward	Displace forks to the left.
	Forward	Displace forks to the right

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Rotational function	Rearward	Rotate frame and body assembly to the left as viewed by the operator.
	Forward	Rotate frame and body assembly to the right as viewed by the operator.

3.10.9 Fire extinguisher. Each truck shall be equipped with a 4 pound internally operated dry chemical charged fire extinguisher with rechargeable cylinder having a minimum UL rating of 6BC. The extinguisher shall be mounted on the truck as shown on the applicable drawing.

3.10.10 Tow hook. A pintle hook assembly as shown on applicable drawing shall be provided at the rear center of each truck securely connected to the counterweight. The pintle and related structure shall be capable of withstanding a longitudinal pull of 34,000 pounds and a lateral pull of 17,000 pounds without permanent deformation or failure when tested as specified in 4.6.19.

3.10.11 Overhead safety guard. The truck shall be provided with a removable overhead guard over the operator's compartment conforming to the applicable drawing. The guard shall be not less than 39 inches above the top of truck seat and vertical braces shall not interfere with operator's front view. Special tools shall not be required to mount and remove guard from truck. The guard shall be capable of withstanding, without permanent deformation or failure, a repeated impact force of 100 pounds when tested as specified in 4.6.18.

3.11 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference limits of the truck shall be in accordance with the above deck and shore requirements of MIL-I-16910.

3.12. Dimensions. The completely fabricated truck shall meet the dimensional limitations specified in Table II and as shown on the drawings listed in section 2.

3.13 Weight. The curb weight shall be the weight of the truck with all hydraulic systems filled to capacity, with coolants and lubricants and with fuel tank one-half (1/2) full. The curb weight shall not exceed 18,500 pounds.

3.14 Finish.

3.14.1 Treatment and painting. All exterior and interior metal surfaces of the truck except the engine and accessories mounted thereon which are normally painted shall be treated and painted in accordance with the provisions of type A of MIL-T-704. Treatment and painting, except color of finish coat, of major sub-assemblies, i.e. engine, transmission and axles, may be done in accordance with the contractor's normal commercial practice if that treatment and painting is compatible with the final finish coat. Unless otherwise specified

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(see 6.2), the final enamel finish coat shall conform to TT-E-485 and shall be Marine Corps semi-gloss green No. 23. Interior surfaces of the hydraulic reservoir and fuel tank shall be preserved in accordance with the suppliers normal commercial practice to compensate for interim storage and then shall be treated with P-20 preservative oil and sealed as specified in 5.1.1.7.5.2. After assembly of the truck, the tanks shall be preserved as specified in Section 5.

3.14.2 Plating. Unless otherwise specified on the drawings listed in Section 2, all threaded fasteners, washers, and cotter pins required to fabricate the truck except those designed to be in contact with engine, transmission, and hydraulic oils shall be cadmium plated in accordance with class 3, type II of QQ-P-416.

3.15 Marking for identification. The entire truck shall be free of all attached, appended or painted markings, decalcomania, name and data plates except as specified herein. Equipment, assemblies and parts shall be identified in accordance with MIL-STD-130 and as shown on the applicable drawings listed in Section 2. The equipment name plate shall be in accordance with MIL-P-514 and be permanently affixed as shown on the applicable drawing.

3.15.1 Instruction and data plates. The truck shall be provided with equipment operational data plates and name plates conforming to the applicable drawings, except as otherwise specified herein. Data plates of the same material as name plates shall be provided by the truck manufacturer describing special operating and servicing procedures including warning and caution instructions, and shall be located in a suitable conspicuous location. Whenever the instruction plates specified in the applicable drawings, are not applicable to the operating procedures of the truck provided with allowable aternate components, the manufacturer shall redesign the plates to include correct data. The redesigned plates shall be subject to approval by the contracting officer and reproducible drawings in accordance with 3.2 shall be furnished.

3.15.2 Registration data. The manufacturer shall execute registration data cards as required by the contracting office (see 6.2). Unless otherwise specified (see 6.2), the assigned registration number for each truck shall be painted, by the contractor, in white lusterless enamel in block numbers 3 inches high on each side of the outside center of the engine hood. The enamel shall conform to TT-E-527 and the color shall be white lusterless number 37875 conforming to Fed. Std. No. 595, unless otherwise specified (see 6.2).

3.16 Government furnished property.

3.16.1 Engine. Unless otherwise specified (see 6.2), the diesel engine shall be Government furnished property.

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3.16.2 Fire extinguisher. Unless otherwise specified (see 6.2), the fire extinguisher specified in 3.10.9 shall be Government furnished property and shall be mounted on the truck as shown on the applicable drawing.

3.16.3 Rifle case. When specified (see 6.2), a Government-furnished rifle case and holder shall be mounted on the truck in a location which is accessible to the operator. The proposed location and method of mounting shall be subject to the approval of the contracting officer prior to installation. The case will be furnished by the Government but fasteners and fittings will be furnished by the supplier.

3.17 Repair parts and maintenance tools. Such repair parts and maintenance tools as specified shall be furnished (see 6.2).

3.18 Items to be furnished with truck. No shipment of trucks shall be made unless repair parts, maintenance tools, operating tools, and accessories as required, are included with the shipment or unless approval for shipment of the trucks without such items has been received from the contracting officer.

3.18.1 Operating and maintenance manual. Unless otherwise specified (see 6.2), the manufacturer shall place one copy of the operating and maintenance manual with each truck at the time of shipment. Manuals shall be complete in every detail in accordance with good automotive practice and should conform to the requirements of the procuring activity. The use of advertising of any type in this manual shall be avoided to maximum extent practicable. Additional copies of instruction manuals and other data requirements shall be furnished as indicated in the contract or order.

3.19 Repair parts support. Unless otherwise specified (see 6.2), contractor will be responsible for making available parts that are peculiar to the contractor or produced exclusively for him. A list of these parts shall be furnished the cognizant technical activity through the contracting officer as a part of the provisioning data to be submitted. These parts shall be available for a period of no less than ten years from date of contract. Should the contractor not make these parts available, he shall furnish the cognizant technical activity sufficient production design and manufacture processing data to facilitate procurement from another source. The contractor shall furnish a certificate that indicates compliance with the foregoing.

3.19.1 Provisioning. Unless otherwise specified (see 6.2), the Government in conjunction with the contractor will as a part of provisioning, develop and establish anticipated usage and pertinent maintenance data of parts peculiar or produced exclusively for him.

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3.20 Workmanship. All workmanship on the truck shall be in accordance with the engineering and production standards of the off-road vehicle industry. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist. Pivot points shall have hardened pins and bushed holes.

3.20.1 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having eroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, and over-heating shall be avoided in accordance with the recommendations of the metal manufacturer. All bends of a major character and all modular assembly fabrication shall be made with metal dies and jig fixtures to insure uniformity of size and shape.

3.20.2 Bolted connections. Bolt holes shall be accurately punched or drilled. Washers and lockwashers shall be provided with the lockwashers required on all bolts, studs, and capscrews having straight threads, unless omission is approved by the contracting officer. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly torqued and shall have full thread engagement.

3.20.3 Welding. Welding shall be in accordance with MIL-STD-1261 and as specified herein and in the drawings listed in Section 2.

3.20.3.1 Welding procedures. Welding procedures shall be in accordance with MIL-STD-1261, except that the procedures required shall be retained by the supplier for use by the Government. Isometric or perspective drawings, showing the location of each joint and the joint welding procedures are not required where shop drawings and written procedures are available.

3.20.3.2 Welding practices. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer shall be thoroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. All weldments shall be free from slag, flux, weld spatter and other impurities. Work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any

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weldment. Butt type joints having members of equal thickness shall be aligned within 10 percent of the thickness of the members involved. Where different steels are welded to each other, filler metal having strength of the lower strength steel shall be used. When the same steels, or steels having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment.

3.20.3.3 Mechanical properties. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification.

3.20.3.4 Soundness. Weld joints of class 2 and class 3 shall be inspected for soundness. Prior to production, fabrication of any weldment of class 2 or class 3 steels, the supplier shall determine the inspection medium (magnetic particle, radiographic, or dye penetrant) necessary to determine the soundness of production welds, and shall develop the information (as applicable) as specified in MIL-STD-1261 for inspection for soundness. This information shall be retained by the supplier and shall be made available to the Government upon request.

3.20.3.5 Welders. Before assigning any welder to work covered by this specification, the contractor shall provide the Government inspection service with names of welders to be employed in the work, together with certification that each of these welders has passed qualification tests as prescribed by either of the following codes for the type of welding operation to be performed and that such qualification is effective as defined by that particular code:

(a) Standard Qualification Procedure of the American Welding Society.

(b) Welding Qualifications of the American Society of Mechanical Engineers.

3.20.4 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended load.

3.20.5 Machine work. All parts shall be manufactured to gage through the use of correct jigs and fixtures. Like parts shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings.

4.1.2 Disassembly inspection. Parts and components failing any test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection.

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5.1)
(d) Inspection comparison (see 4.5.2).
(e) Inspection of preparation for delivery (see 4.5.3).

4.3 Preproduction inspection. The preproduction model specified in 3.2 shall be designated truck-A.

4.3.1 Preproduction examination.

4.3.1.1 Pretest examination. Truck-A shall be examined for the defects marked "X" in column 1 of table IV. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post test examination. Upon successful completion of all tests specified in 4.3.2, the preproduction truck shall be examined for the defects marked "X" in column 2 of table IV. Presence of one or more defects shall be cause for rejection.

TABLE IV. Examination Schedule

Pretest	Post-Test	Ini. Prod	Individ.	Sample	Defects	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	-	X	101. Dimensions and design not as shown on the Government drawings.	3.1 + 3.12

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Table IV. Examination Schedule (Cont'd)

Pretest	Post-test	Initial Prod.	Individual	Sample	Defects	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	-	-	102. Government furnished seals not installed as specified.	3.2
X	-	X	-	X	103. Materials not as specified.	3.4
X	-	X	-	X	104. Threads not in accordance with Handbook H28.	3.4.2.1
X	-	X	-	X	105. Truck dimensions and weights not as specified.	Table II 3.1.3 + 3.1.4
X	-	X	-	X	106. Fork dimensions not as specified.	Table II + 3.10.8.1
X	-	X	-	X	107. Fuel tanks not as specified.	3.10.2.1
X	-	X	-	X	108. Vent tubes and pipes not extended above 5 foot fording depth.	3.9.2
X	-	X	-	X	109 Exhaust system not as specified.	3.10.3
X	-	X	-	X	110. Power train components not as specified.	3.10.5
X	-	X	-	X	111. Transmission not as specified.	3.10.5.1
X	-	X	-	X	112. Transmission accessories not as specified.	3.10.5.1
X	-	X	X	X	113. Transmission controls do not operate as specified.	3.10.5.1.1
X	-	X	-	X	114. Interaxle differential not as specified.	3.10.5.1.2
X	-	X	-	X	115. Universal joints and drive shafts not as specified.	3.10.5.2
X	-	X	-	X	116. Front and rear axles not as specified.	3.10.5.3

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Table IV. Examination Schedule (Cont'd)

Pretest	Post-test	Initial Prod.	Individual	Sample	Defects	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	-	X	117. Axle oscillation not as specified.	3.10.5.3.1
X	-	X	-	X	118. Tires, rims and components not as specified.	3.10.5.4
X	-	X	-	X	119. Overhead guard not as specified.	3.10.12
-	-	-	X	-	120. Any rubber product over age.	3.4
X	-	X	-	X	121. Steering system not as specified.	3.10.6
X	-	X	-	X	122. Brake system not as specified.	3.10.7
X	-	X	X	-	123. Parking brake rough, chatters, or not as specified.	3.10.7.2
X	-	X	-	X	124. Engine not as specified.	3.10.1
X	-	X	X	-	125. Engine data plat not as specified.	3.10.1(g)
X	-	X	-	X	126. Electrical system not as specified.	3.10.4
X	X	X	X	-	127. Electrical wiring not protected from chafing as specified	3.10.4
X	-	X	-	X	128. Hydraulic system not as specified.	3.10.8
X	-	X	X	-	129. Hydraulic controls do not operate as specified.	3.10.8.5
-	-	X	X	-	130. Electro-magnetic interference reduction not as specified.	3.11
X	-	X	X	-	131. Fuel, lubricants, and coolant not as specified.	3.7
X	X	X	X	X	132. Leaks in coolant, fuel, oil, and air lines.	Section 3

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Table IV. Examination Schedule (Cont'd)

Pretest	Post-test	Initial Prod.	Individual	Sample	Defects		Requirement Paragraph
1	2	3	4	5	6		7
X	-	X	-	X	133.	Lubrication fittings not as specified.	3.7.1 + 3.7.2
X	X	X	X	X	134.	Seals do not prevent entrance of foreign material or leakage of fluid or grease	Section 3
X	-	X	X	X	135.	Identification marking not as specified.	3.15
X	-	X	X	X	136.	Cleaning, treatment, painting, and marking not as specified	3.2.2.3
-	-	-	X	-	137.	Assembly incorrect or not complete	Section 3
-	-	-	X	-	138.	Parts or components missing or do not function.	Section 3
-	-	-	X	-	139.	Material not of same quality as used on preproduction model.	3.2
X	-	X	X	-	140.	Workmanship not as specified.	3.2

TABLE V. TEST SCHEDULE

				Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	X	X	Test conditions	4.6.3	-
X	-	X	-	Fuel tank test	4.6.4.3	3.10.2.1
X	X	-	-	Hydraulic systems (Test method 1)	4.6.5	3.10.8 & 3.20

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Table V. Test Schedule

Prepro Model A	Initial Production	Individual	Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
-	-	X	-	Hydraulic systems (Test method 1, paragraph 2a, b and c only)	4.6.5	3.10.8 & 3.20
-	-	X	-	Run-in	4.6.6	3.20
X	X	-	-	Travel speed test (Test method 2)	4.6.7	3.9.6
X	X	-	X	Acceleration	4.6.8	3.9.7
X	X	-	-	Slope operation (Test method 3)	4.6.9.1	3.9.4 & 3.10.7.1
X	-	-	X	Weight Distribution	4.6.9.2	3.9.3 & 3.13
X	-	-	-	Torque converter stall	4.6.10.1	3.10.5.1
X	-	-	-	Axle differential	4.6.10.2	3.10.5.3
X	-	-	-	Tire test	4.6.10.3 & 4.6.22.2	3.10.5.4
X	-	-	-	Low temperature (-25°F) (Test method 4)	4.6.11	3.6.1
X	-	-	-	High temperature (120°F) (Test method 5)	4.6.12	3.6.1
X	-	-	-	Lifting attachments (Test method 6)	4.6.13	3.8
X	X	-	X	Steering system tests	4.6.14	3.10.6.1 thru 3.10.6.3
X	X	-	X	Brake tests	4.6.15	3.10.7
X	X	-	-	Hydraulic function and forklift mechanism test	4.6.16	3.10.8
-	-	-	X	Forklift mechanism and body rotation test	4.6.16.1	3.10.8 & 3.9.1

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Table V. Test Schedule (Cont'd)

Preprod Model A	Initial Production	Individual	Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	-	-	Fork rack test	4.6.17	3.10.8.1
X	-	-	-	Overhead safety guard	4.6.18	3.10.11
X	-	-	-	Tow hook test	4.6.19	3.10.10
X	-	-	-	Electromagnetic interference	4.6.20	3.11
X	X	X	-	Lighting	4.6.6 & 4.6.21	3.10.4
X	-	-	-	Endurance test (Load placement) (Test method 9)	4.6.22.1(a)	3.10.8.3 & 3.9.5
X	X	-	-	Endurance test (Durability) (Test method 8)	4.6.22.1(b)	3.10.8.3 & 3.9.5
X	-	-	-	Endurance test (Sea water) (Test method 10)	4.6.22.2	3.9.2
X	-	-	-	Post test disassembly and examination	4.6.23	Section 3

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, truck-A shall be subjected to the tests marked "X" in column 1 of table V. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. An initial production truck will be selected at random by the Government from the first 10 trucks being produced from production tooling and will be examined for the defects marked "X" in column 3 of table IV and subjected to the tests marked "X" in column 2 of table V, to determine conformance to the requirements of this specification. Acceptance of the initial production truck shall not exclude the remaining trucks from the quality

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conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of the initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on truck previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all trucks accepted prior to the completion of initial production tests are similarly deficien unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Production inspection.

4.5.1 Quality conformance inspection.

4.5.1.1 Individual examination. After successful completion of all tests specified in 4.5.1.2 each truck shall be examined for the defects marked "X" in column 4 of table IV. Presence of one or more defects shall be cause for rejection.

4.5.1.2 Individual tests. Each truck shall be subjected to the tests marked "X" in column 3 of table V. Failure of any tests shall be cause for performing the inspection specified in 4.1.2.

4.5.1.3 Sampling.

4.5.1.3.1 Unit of product. A unit of product is defined as a fully serviced truck.

4.5.1.3.2 Lot. A lot shall consist of not more than 50 units of product. Lots shall be formulated from trucks which have successfully completed the individual examination and individual tests as specified in 4.5.1.1 and 4.5.1.2 respectively.

4.5.1.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the tests marked "X" in column 4 of table V. Any sample failing any test shall be cause for rejection of the lot.

4.5.1.3.4 Sample examination. After successful completion of the tests as specified in 4.5.1.3.3, each sample shall be examined for the defects marked "X" in column 5 of table IV. Presence of one or more defects shall be cause for rejection of the lot.

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4.5.2 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination specified in columns 1 and 4 of table IV and to the tests specified in column 1 of table V to reveal any manufacturing deficiencies. The inspection will be performed by the Government at a site selected by the Government. Trucks will be selected at random from the previously selected preproduction models, the initial production model and those trucks tested as 4.5.1.3 Rehabilitation shall be at Government expense.

4.5.2.1 Inspection failure. Failure of any truck when examined and tested as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of trucks until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.5.3 Inspection of preparation for delivery.

4.5.3.1 Quality conformance inspection of pack.

4.5.3.1.1 Unit of product. For the purpose of inspection, a completely processed mobile truck prepared for shipment shall be considered a unit of product.

4.5.3.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.3.1.3 Examination. Samples selected in accordance with 4.5.3.1.2 shall be examined for the following defects (AQL shall be 2.5 percent defective):

150. Materials, methods, and containers not as specified for level A. Each incorrect material or incorrect method or incorrect container shall constitute a defect.

151. Items not disassembled from truck as specified.

152. Matchmarking not as specified.

153. Tags used for matchmarking not waterproofed as specified.

154. Openings into transmission and torque converter housing not sealed with tape and controls not tagged as specified.

155. Exposed surfaces of shaft, splines, slip joints, steering knuckles, and universal joints not coated with specified preservative.

156. Hydraulic ramshafts not coated with preservative, surfaces not wrapped, and ramshafts not tagged as specified.

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157. Hydraulic controls not tagged as specified.

158. Hydraulic controls not secured in neutral position.

159. Engine cooling system not preserved as specified, drain cocks not left open, and system not tagged as specified.

160. Engine lubrication system not preserved and system not tagged as specified.

161. Engine combustion chambers and valves not preserved as specified.

162. Warning tag "Engine preserved - Do not crank until issued to user" not attached in a conspicuous location within the driver's compartment.

163. Openings into exhaust systems including removed stacks not sealed with tape as specified.

164. Tension on drive belts not released as specified, and pulleys not tagged as specified.

165. Unpainted surfaces of pulley grooves not coated with primer as specified.

166. Openings into air cleaner not sealed with tape as specified, and air cleaner not tagged as specified.

167. Battery cables not disconnected and not secured within the box.

168. Vent openings in batteries not moisture sealed as specified and battery not tagged as specified.

169. Roller chains and sprockets not preserved as specified.

170. Openings into electrical components not sealed with tape as specified, and components not tagged as specified.

171. Components and surfaces of booms not coated with specified preservatives.

172. Unpainted surfaces of rollers not coated with preservative as specified.

173. Exterior surfaces and components of brakes not coated with preservative as specified.

174. Pneumatic tire pressure not adjusted as applicable, as specified.

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175. Wood blocks not placed between axles and frame to prevent oscillation and excessive bouncing.

176. Wheel hubs not tagged as specified.

177. All operating levers not secured in neutral position, control knobs not removed and lever threads not coated with preservative.

178. Forks not repositioned on the truck as specified.

179. Strapping used to secure forks to rack not class B.

180. Wood blocks or dunnage not used when required to prevent loose straps as specified.

181. Overhead guard not placed and not secured in the transport position.

182. Pins removed from overhead guard not coated with preservative and not reinstalled.

183. Seat backs, cushions, and seat belts not protected as specified.

184. Strapping on intermediate containers not class B.

185. Intermediate containers not positioned and secured on the truck as specified.

186. Warning tag not attached to steering wheel to caution against turning wheel as specified.

187. Warning tags and printing not waterproofed.

188. Marking illegible, incomplete, incorrect.

189. Depreservation guide not properly prepared, not placed in waterproof envelope, and not attached as specified.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table IV.

4.6.2 Tests. Tests shall be conducted in accordance with the test schedule in Table V in the sequence shown and as required by 4.2.

4.6.3 Test conditions. Prior to test, the truck shall be lubricated with oils and greases specified in 3.7. Oils shall be those

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designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. Evidence of non-compliance with the requirements of this specification shall be cause for rejection in the tests specified below.

4.6.4 Preassembly test.

4.6.4.1 Fuel tank test. Prior to application of finish each fuel tank shall be tested to determine that there is no leakage. A dye penetrant shall be added to a noncombustible fluid having the same viscosity as diesel fuel. The tank shall be pressurized by air to 4 psig and visual inspection for leaks shall be made. Except for the opening used to apply air pressure, all openings (drain, vent, fill cap, et cetera) shall be closed. Any evidence of leakage shall constitute failure of this test.

4.6.5 Hydraulic system. The hydraulic system shall be tested in accordance with test method 1. (See appendix for all test methods.) Evidence of contamination, a draft rate exceeding 1 inch per hour, malfunction or bending of any cylinder, evidence of leakage that cannot be stopped as specified in 2d of the test method, any cavitation, or any reservoir leakage shall constitute failure of this test. The individual testing of each truck shall be limited to 2a,b and c of test method 1.

4.6.6 Run-in. The truck after completion of 4.6.5 shall be run-in for one hour on the 25 foot diameter circle in accordance with test method No. 9 (load placement test) and for 10 cycles through the durability course at 3.5 miles per hour in accordance with test method No. 8. During operations and after completion of the run-in, the truck shall be examined for leaks, operational failures, failure of any component to perform as specified, or permanent deformation. Any evidence of the above shall constitute failure of the test.

4.6.7 Travel speed. The travel speeds of the truck shall be determined in accordance with test method No. 2. Speeds not in accordance with 3.9.6 shall constitute failure of this test.

4.6.8 Acceleration. The unload truck shall be driven over a paved, dry, level 100-foot course to determine compliance with 3.9.7. The resultant time for this test shall be determined by averaging 5 consecutive test runs and truck shall be in a stopped position at the beginning of the course. Noncompliance with the requirements above shall constitute failure of this test.

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4.6.9 Stability.

4.6.9.1 Slope operation. The trucl shall be tested in accordance with test method No. 3. Inability to negotiate the slope, evidence of any wheel or wheels leaving the ground, inability to stop and restart the truck when on the slope, inability of the brake holding device to hold the trucks, evidence of any leaks, or lack of lubrication shall constitute failure of this test.

4.6.9.2 Weight distribution. The truck and its wheel loads shall be weighed to determine compliance with 3.9.3 and 3.13. Evidence of non-compliance shall be cause for rejection.

4.6.10 Power train tests.

4.6.10.1 Torque converter stall. With the service brakes fully applied and the torque converter oil at a temperature of 180°F. measured at the converter outlet to the cooler, engage the transmission and run the engine at wide open throttle for a period of not less than 30 seconds. Record the temperature at the converter outlet. A temperature in excess of 250°F. shall constitute failure of this test.

4.6.10.2 Axle differential. The truck shall be positioned on a level surface with three wheels on a greased area to allow slippage of the wheels. The truck shall be driven by one wheel with the transmission in its lowest gear to determine compliance with 3.10.5.3. Rated load shall be placed on forks in carry position for this test. The test shall be repeated with all 4 wheels placed on a greased area to insure that no movement is possible under greased conditions.

4.6.10.3 Tire test. The truck with rated load in carry position shall be operated over soft sand, deep snow or mud at a minimum speed of 2 mph for a period of 30 minutes to determine compliance with 3.10.5.4. Evidence of non-compliance shall be cause for rejection.

4.6.11 Low temperature (minus 25°F.) The truck shall be tested in accordance with test method No. 4. One or more of the following shall constitute failure of this test:

(a) Engine does not start within 5 minutes after beginning the first cranking.

(b) Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of primers or priming aids.

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(c) Lubricating-oil temperature does not stablize in the engine crankcase within the range of 130°F. to 230°F. and remain within this range throughout the period of engine cycling.

(d) Coolant temperature does not stablize in the engine-cooling jacket discharge pipe within the range of 140°F. to 210°F. and remain within this range throughout the period of engine cycling.

(e) Instruments and gages do not operate.

(f) Battery-charging system does not show positive charge.

(g) With the operator wearing arctic mittens and boots:

(1) Power train cannot be engaged to the wheels.

(2) Any clutch, brakes, shaft, drum, or cylinder cannot be operated through two complete movement cycles.

4.6.12 High temperature (plus 120°F.). The truck shall be tested in accordance with test method No. 5. Inability to start the engine within 5 minutes, inability to operate the engine under rated load conditions within 15 minutes without unnatural or continued control manipulation, inability to maintain engine temperatures within the range specified by the engine supplier, transmission, torque converter or transfer case oil temperature in excess of 250°F., hydraulic system oil temperature in excess of 225°F., or failure of controls and instruments to operate after 15 minutes shall constitute failure of this test.

4.6.13 Lifting attachment. The truck shall be tested in accordance with test method No. 6. Weld failure or permanent deformation of the truck or its lifting attachments clearance of less than 1 inch between the slings and the truck, or slings converging more than 19 feet above the lowest extremity of the truck shall constitute failure of this test.

4.6.14 Steering system tests.

4.6.14.1 Steering overload. Place truck steering wheel in the extreme position of turn and apply a force of 200 pounds tangent to the periphery of the steering wheel to determine compliance with 3.10.6.1. This test shall be repeated 5 times in the extreme right hand position of turn and 5 times in the extreme left position of turn (with and without utilizing power assist).

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4.6.14.2 Turning radius. With truck wheels turned to extreme right and left turn positions, measure radius of turn to determine compliance with 3.10.6.2.

4.6.14.3 Steering effort. Place unloaded truck in a stationary position on a dry, level, brushed or trowled concrete surface, and rotate steering wheel (utilizing power asist) from extreme right to extreme left position and vice versa by applying a force to the periphery of the steering wheel, to determine compliance with 3.10.6.1.

4.6.14.4 Crabbing angle. The angle between straight forward driving and driving with the wheels in the extreme crab position shall be measured to determine compliance with 3.10.6.3. Suggested method of angle measurement is shown in figure 1.

4.6.15 Brake tests. Brake tests shall consist of all the tests specified in 4.6.15.1 through 4.6.15.5.

4.6.15.1 Service brake effort (slope). The truck with rated load in carry position shall be driven rearward down at 45 percent prepared surface slope at 5 mph and the service brake pedal shall be fully actuated to determine compliance with 3.10.7.1.1(a).

4.6.15.2 Service brake effort (level surface). The truck with rated load in carry position shall be driven at 5 mph on a level, dry paved surface and the service brake pedal fully actuated to determine compliance with 3.10.7.1.1(b). This test shall be repeated 10 times.

4.6.15.3 Service brake effort (after water immersed). The truck with rated load in carry position shall be driven into water of sufficient depth to completely immerse brakes. Truck shall then be positioned on 45 percent slope with service brake pedal fully actuated to determine compliance with 3.10.7.1.1(c). During sample testing, the brake holding device shall then be actuated and the service brakes be fully disengaged to determine compliance with 3.10.7.1.3.

4.6.15.4 Service brake deceleration. The truck with rated load in carry position and forks tilted rearward at 30 degrees shall be driven forward at maximum speed over a level, dry paved surface and a force of 125 pounds shall be applied to the service brake pedal to determine compliance with 3.10.7.1.2.

4.6.15.5 Parking brake effort. The truck with rated load in carrying position shall be driven at 2 mph over a level, dry paved surface and the parking brake shall be fully actuated to determine compliance

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with 3.10.7.2. Transmission shall be in neutral when parking brake is actuated. This test shall be repeated at least 3 times each in forward and reverse directions.

4.6.16 Hydraulic function and forklift mechanism test. The test on the hydraulic forklift mechanism hereinafter specified shall be performed in the sequence listed. Frame and body rotational function specified in 3.10.8 shall be included in cycling test indicated for lift mechanism maneuvers.

4.6.16.1 Forklift mechanism and body rotation maneuver. The truck with rated load on forks, shall be positioned on a level surface and all hydraulic controls shall be operated to determine that lift mechanism function and rotational function can be performed in accordance with requirements specified in 3.10.8 and 3.9.1.

4.6.16.2 Static load test. The truck shall be tested in accordance with test method No. 7. Breakage or evidence of permanent deformation of the truck shall constitute failure of this test.

4.6.16.3 Dynamic load tests. The following dynamic load tests shall be performed and truck shall be examined after completion of tests to determine compliance with 3.10.8.

(a) The truck shall be placed on a level surface and all hydraulic and mechanical control devices shall be set for normal operation. Place 115 percent of rated load on lift forks and subject lift mechanism to 100 cycles of lifting (to maximum height) and lowering tilting to a maximum tilt angle (forward and reverse) and including frame and body rotation to maximum left and right rotation angles for each cycle. Remove load and examine truck. Pressure relief valve shall be operative for this test.

(b) Repeat test "a" except use 150 percent of rated load on forks and perform only 50 cycles. The pressure-relief valve shall be inoperative for this test.

(c) Repeat test "A" except place rated load on lift forks and perform 1,000 cycles. The lowering maneuver for the final 20 cycles of this test shall be performed by abruptly releasing load at maximum height position and abruptly stopping load 12 inches above ground to determine compliance with 3,10.8.3. Prior to removing the load at completion of this test, raise load to maximum height with lift fork in horizontal position and retain load for five munute period. Check the downward drift, longitudinal tilt drift and rotational function drift to determine compliance with 3.10.8.

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4.6.17 Fork rack. With the truck on a level surface, place a uniformly distributed rated load on the rack as shown in figure 2. The fork rack shall be lifted to maximum height and fully tilted to the rear. The load shall be held in this position for 1 minute. Remove the load and examine the rack. Breakage or evidence of permanent deformation of the rack shall constitute failure of this test.

4.6.18 Overhead guard. Test overhead guard to determine compliance with 3.10.11 by dropping a wooden box weighted to 100 pounds on the guard from a height of not less than 8 feet above guard. The base of the box shall be 1 foot square and drop test shall be repeated 10 times.

4.6.19 Tow hook test. Apply a longitudinal pull of 34,000 pounds and a lateral pull of 17,000 pounds to the truck tow hook to determine compliance with 3.10.10. Pull loads shall not be applied at the same time.

4.6.20 Electromagnetic interference. The truck shall be tested in accordance with the applicable provisions of MIL-I-16910 to determine compliance with 3.11.

4.6.21 Lighting. All lights shall be turned on and left on during all endurance tests specified in 4.6.6 and 4.6.22 as applicable. Failure of any electrical component other than light bulbs or sealed lighting units shall constitute failure of this test.

4.6.22 Endurance.

4.6.22.1 Durability, and load placement. The truck shall be tested on a minimum 16-hour-per-day, 5-day-per-week continuous basis (with downtime allowed within the 16-hour-per-day operational time for scheduled servicing) in accordance with test methods Nos. 8 and 9. Hours specified in test methods 8 and 9 shall be productive clock hours excluding fueling, servicing, and downtime. Inability of the truck to accomplish all of the following operations without malfunction, or permanent deformation; or to perform all tests without exceeding specified temperature limits of any component or system shall constitute failure of this test.

(a) To transfer 108 tons per hour (see test method no. 9).

(b) To negotiate the test course not less than 25 times per hour for 300 hours (see test method No. 8).

4.6.22.2 Sea water. The truck shall be tested at a Goverment facility at Government expense (see 6.3) in accordance with test method No. 10. The truck shall be rehabilitated by the contractor when required to insure the waterproof boots and seals are intact prior to testing.

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Upon completion of this test, the truck shall be subjected to the test specified in 4.6.23. Any of the following shall constitute failure of this test:

(a) Evidence of water damage to or malfunction of the torque converter, engine oil sump, transmission, transfer case, fuel tank, hydraulic reservoir, brake reservoir, either axle, any wheel, instruments, or gages.
(b) Malfunctioning of any electrical component.
(c) Malfunctioning of any brake.
(d) Malfunctioning of any hydraulic component.
(e) Malfunctioning of the steering system.
(f) Evidence of corrosion which affects the correct functions of the truck.
(g) Inability to start the engine in the water.

4.6.23 Post test disassembly and examination. Upon completion of all tests, the hydraulic reservoir shall be pressurized and checked for leaks; each engine cylinder compression shall be measured and recorded; and the truck shall be dismantled, and the following components shall be disassembled in detail for visual examination:

(a) Engine - The cylinder head(s), all valves, two pistons and piston pins removed from the two cylinders having the lowest compression pressure reading at the end of the test, two main bearings and two connecting rod bearings removed, and turbo-charger (if used).
(b) Torque converter and transmission.
(c) Universal joints and drive shafts.
(d) Differential carrier (front and rear).
(e) Final drive planetaries (right front and left rear).
(f) Brake system (complete).
(g) Steering system.
    (1) All cylinders.
    (2) All valves.
    (3) Pump.
(h) Hydraulic system.
    (1) Pumps and valves.
    (2) All cylinders
    (3) Controls
    (4) Hose and fittings where leakage has occurred.
(i) Wheels and bearings (right front and left rear).

Additional disassembly and examination shall be made at the option of the contracting officer. A complete description of the condition of all components and parts of the truck shall be recorded, and photographs shall be taken of any deficiency. Each evidence of corrosion which affects the correct functions of the truck shall constitute a defect. Each evidence of permanent deformation, breakage, or excessive wear of any part or component shall constitute a defect. The contracting officer shall have the final determination as to what constitutes excessive wear or corrosion.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be

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level A, B or C, as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Disassembly and matchmarking. The controls, knobs, vertical exhaust stacks, right running light, rear working light, overhead working light, and fire extinguisher shall be removed from the truck. Parts removed and parts remaining on the truck shall be matchmarked to facilitate reassembly. Matchmarking shall be on cloth tags conforming to UU-T-81, type A, and the tags attached to mating parts. Tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.2 Cleaning and drying. Surfaces to which a preservative is to be applied shall be cleaned and dried in accordance with MIL-P-116, cleaning process C-1 and drying procedure D-1 through D-4.

5.1.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in acoordance with MIL-P-116.

5.1.1.4 Methods of preservation. Methods of preservation indicated by symbols herein shall conform to the corresponding symbols listed in MIL-P-116.

5.1.1.5 Power trains.

5.1.1.5.1 Automatic transmission and torque converter. The automatic transmission and torque converter housing shall be filled to the operating level with lubricating oil conforming to MIL-L-2104, grade 10. The transmission and torque converter shall be operated through all stages for a minimum of 1 minute at sufficient engine speed to assure lubricant coverage of all interior parts and surfaces with the lubricant. Seal openings into housing with tape conforming to PPP-T-60, class 1. A warning tag shall be attached to the controls indicating: "The housing is filled to the operating level with the approved lubricant required for operation - Do not drain. Remove all seals from housing openings." The tags shall conform to UU-T-81, type A. The tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.5.2 Differential, transfer cases, and final drives. Differential, transfer cases, and final drive housing shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade 90, and operated as specified in 5.1.1.5.1. A warning tag shall be attached to the operating controls indicating: "The housings are filled to the operating level with the approved lubricant required for operation - Do not drain." The tags shall conform to UU-T-81, type A. The tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

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5.1.1.5.3 Shafts and steering knuckles. Exposed surfaces of shafts, including splines, slip joints, and universal joints shall be coated with type P-1 preservative. The steering knuckles shall be sprayed with type P-7 preservative.

5.1.1.6 Hydraulic systems. The hydraulic control systems shall be filled to the operating level with lubricating oil conforming to MIL-L-2104, grade 10. Lift and tilt assemblies shall be operated to full extent to insure coating of all interior surfaces. A warning tag shall be attached to the hydraulic controls indicating: "The hydraulic system is filled to the operating level with the approved fluid required for operation - Do not drain." The tag shall conform to UU-T-81, type A. Hydraulic ramshafts shall be retracted to transport position. The hydraulic controls shall be secured in the neutral position. Any exposed surfaces of the shafts shall be coated with type P-6 preservative and the coated surfaces wrapped with greaseproof barrier material conforming to MIL-B-121, type 1, grade A, class 1, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, class 1. A tag shall be attached to each ramshaft indicating: "Remove tape and barrier material prior to activating the systems." The tags shall conform to UU-T-81, type A. Tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.7 Engine.

5.1.1.7.1 Cooling system. Prior to start of engine preservation, the cooling system shall be filled with type P-3 preservative. The preservative shall remain in the system during the entire engine preservation and then shall be drained. After draining, the drain cocks shall be left open. A warning tag shall be attached to filler neck indicating: "Close drain cocks and fill cooling system before operating the engine:" The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129. The tag shall be securely attached to the filler neck.

5.1.1.7.2 Lubrication system. Prior to start of engine preservation, the engine crankcase shall be filled to the operating level with type P-10, grade 2 preservative. The preservative oil shall remain in the crankcase during entire engine preservation and for shipment and storage of the truck. Upon completion of engine preservation a warning tag shall be attached to the crankcase dipstick indicating: "The crankcase is filled with type P-10 preservative lubricating oil and the preservative lubricating oil shall be drained and the crankcase filled to the operating level with the approved lubricating oil (MIL-L-2104, grade as applicable), prior to operating the engine." The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

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5.1.1.7.3 Fuel system. The engine fuel intake line shall be disconnected at the most easily accessible point nearest the fuel tank, and a line from a portable container filled with diesel fuel conforming to VV-F-800 shall be connected to the line leading to the engine. Injector fuel return line shall be disconnected at the quick disconnect coupling. A line shall be provided and connected to the injector fuel return quick disconnect coupling to permit draining onto a recovery container. The engine shall be started and operated at fast idle until thoroughly warm. The engine shall then be accelerated to 3/4 speed, at the same time the fuel supply shall be switched to a portable container containing type P-9 preservative oil. The engine shall be operated until the entire fuel system is filled with the preservative oil. The engine shall be stopped and the fuel lines reconnected. (The recovered fuel-oil mixture shall not be used for preserving other engines.) After the engine has cooled to a maximum cylinder head temperature of 100°F. the combustion chambers and valves shall be preserved as specified in 5.1.1.7.4.

5.1.1.7.4 Combustion chambers and valves. The diesel throttle shall be completely closed. The air box covers on the side of the engine opposite the blower shall be removed. The exhaust manifold and rocker arm cover shall be removed. The engine crankshaft shall be rotated with the starting motor until the piston in the cylinder to be atomized sprayed is below the intake port. While the intake port into the cylinder is open, .5 ounce of type P-10, grade 2, preservative shall be sprayed into the open port. The same preservation shall be performed on each cylinder until all cylinders have been atomized sprayed with the specified amount of preservative oil, after which the crankcase shall not be rotated. With each exhaust valve manually depressed, .5 ounce of type P-10, grade 2 preservative shall be atomized sprayed through the exhaust port. The interior of the air box cover shall be coated with the type P-10, grade 2 preservative oil. Rocker arm assemblies, valves springs, and the interior of the rocker arm cover shall be atomized sprayed with the type P-10, grade 2 preservative. The air box cover, exhaust manifold, and the rocker arm cover along with new gasket for each shall be reinstalled. A warning tag indicating: "Engine has been preserved - Do not crank until issued to the user" shall be securely attached in a conspicuous location within the operator's compartment. The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.7.5 Fuel tanks. Uncoated fuel tanks shall be preserved in accordance with one of the procedures specified in 5.1.1.7.5.1 or 5.1.1.7.5.2.

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5.1.1.7.5.1 Preservative and drain method. The fuel tank shall be completely drained of fuel and then completely filled with type P-10, grade 2 preservative, and then again drained. The tank shall be allowed to drain until the oil flow ceases. The drain plug shall then be coated with type P-10, grade 2 preservative oil and the plug reinstalled. The vent hole in the filler cap shall be sealed with tape conforming to PPP-T-60, class 1. A tag shall be attached to the filler neck indicating: "Remove tape from fuel vent prior to operating the engine." The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129. The drained preservative oil may be used for processing other fuel tanks provided not more than 10 percent of the fluid is diesel fuel.

5.1.1.7.5.2 VCI fill and drain method. The fuel tank shall be preserved and tagged as specified in 5.1.1.7.5.1, except that P-20 shall be used in lieu of type P-10, grade 2 preservative oil.

5.1.1.7.6 Exhaust system. Openings into exhaust systems including exhaust stack openings when removed, shall be completely sealed with tape conforming to PPP-T-60, class 1. Vertical exhaust stacks, when removed, shall be placed and secured to the truck behind the operator's seat. Any exterior uncoated surfaces of the stacks and muffler shall be coated with type P-1 preservative. Clamps shall be coated with type P-1 preservative and reinstalled in place.

5.1.1.7.7 Drive belts. Tension of all drive belts shall be released. Unpainted surfaces of pulley grooves shall be coated with primer conforming to TT-P-664. A warning tag indicating: "Belt tension released - Adjust to engage accessories", shall be attached in a conspicuous location on each pulley. The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.7.8 Air cleaner. The air cleaner shall remain in place on the truck. Openings into the air cleaner shall be sealed with tape conforming to PPP-T-60, class 1. A warning tag shall be attached indicating: "Remove tape from openings prior to operating the engine." The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.7.9 Batteries, battery cables, and electrolyte. Battery cable shall be disconnected from the batteries. The cable ends shall be secured within the battery box with tape conforming to PPP-T-60, class 1. The batteries shall remain in place in the battery box. The vent openings shall be sealed by placing a high density film (polyethylene film, 1 mil. thick or polyester tape) over the fill openings and the fill caps turned into place over the film. The required amount of electrolyte shall be packaged in type IV units in accordance with MIL-S-207, and overpacked in a wood container in accordance with MIL-S-207. The electrolyte pack

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shall be secured to the floor of the operator's compartment. A warning tag shall be attached indicating: "Remove moisture seals under fill caps, reconnect battery cable, and fill the battery with the required electrolyte." The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.8 Roller chains and sprockets. Coat roller chains with enough type P-3 or type P-9 preservative to assure penetration to all inner surfaces of the rollers, pins, and bushings. After excess preservation has drained, the entire chain and sprockets shall be coated with type P-1 preservative.

5.1.1.9 Electrical components. Openings into electrical components shall be sealed with tape conforming to PPP-T-60, class 1. A tag shall be attached to each component indicating: "Remove tape from openings prior to operating the truck." The tags shall conform to UU-T-81, type A. The tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.10 Boom. Coat moving parts with type P-11 preservative. Coat other exposed unpainted surfaces with type P-1 preservative.

5.1.1.11 Rollers. Coat all exterior unpainted surfaces of rollers with type P-1 preservative.

5.1.1.12 Brakes. Exterior unpainted or threaded surfaces of cables, clevises, and linkages shall be coated with type P-1 preservative. The brakes shall be actuated to insure coating of all interior surfaces with the operating fluid.

5.1.1.13 Pneumatic tires. Pneumatic tires (road tires), shall be inflated to 10 pounds above pressure value recommended. Tires mounted on spare rims shall be deflated to two-thirds of the pressure value recommended. To prevent oscillation and excessive bouncing on the tires, wood blocks shall be placed between the axles and the frame. A warning tag shall be attached to each wheel hub indicating: "Adjust pressure value of tires and remove wood blocks between axles and frame." The tags shall conform to UU-T-81, type A. The tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.1.1.14 Operating levers. All operating levers shall be secured in neutral position. The control knobs shall be removed and the control lever threads coated with type P-1 preservative. The knobs shall be placed in the tool box.

5.1.1.15 Forks. The forks and fork rack shall be retracted to the transport position reducing overall cubage to a minimum.

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5.1.1.16 Overhead guard. The overhead guard shall be placed and secured in the transport position. Pins removed shall be coated with type P-1 preservative and reinstalled.

5.1.1.17 Running lights. Each running lifht removed from the truck shall be packaged in a close fitting box conforming to PPP-B-636, W5c or W6c, style RSC. After closure the box shall be waterproofed sealed with tape as specified for interior containers in accordance with the appendix to the box specification.

5.1.1.18 Fire extinguisher. When not Government furnished property, each fire extinguisher shall be packaged in a box as specified in 5.1.1.17.

5.1.1.19 Seat backs, cushions, and seat belts. Seat backs, cushions, and seat belts shall remain in place and shall be protected by completely covering with a shroud of black polyethylene conforming to L-P-378, minimum 6 mil. thickness. The shroud shall be secured in place with tape conforming to PPP-T-60, class 1.

5.1.1.20 Repair parts. The preservative application criteria and applicable submethod(s) of MIL-P-116 shall be used to preserve and package the repair parts.

5.1.1.21 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of MIL-H-15424, and placed in the tool box.

5.1.1.22 Lubrication order, log book, and technical manuals. Preserve the technical publications together, method IC-1. The preserved publications shall be placed in the tool box.

5.1.1.23 Intermediate packaging. The running lights, fire extinguisher, and repair parts shall be intermediate packaged together in a box conforming to PPP-B-601, overseas type, style A or B. Strapping shall be class B.

5.1.2 Level C. The truck, components, repair parts, tools and publications shall be preserved and packaged as specified herein to afford protection against deterioration and damage from the supplier to the initial destination for immediate use.

5.1.2.1 Batteries and electrolyte. Wet batteries shall be secured in place in vehicle battery carrier. When dry charge batteries are furnished, batteries, cabless and electrolyte shall be protected as specified in 5.1.1.7.9 as applicable.

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5.1.2.2 Cooling system. Cooling systems shall be protected by the water and antifreeze procedure specified in 5.1.1.7.1.

5.1.2.3 Engines, hydraulic systems, power train components.

5.1.2.3.1 Engines. With the engine cooled to a cylinder head temperature of not more than 100°F., completely close the throttle and remove the covers and plugs or flanges from the openings. Type P-10, grade 2, preservative oil shall then be atomized sprayed into each combustion chamber and cylinder through the openings. The amount of type P-10 shall be as specified in 5.1.1.7.4. Rotate the engine not less than 5 complete revolutions while each cylinder is being sprayed. Coat the interior surfaces of the covers, plugs, and flanges, with type P-10, grade 2, preservative oil and reinstall.

5.1.2.3.2 Hydraulic systems, power train components. Hydraulic systems and power train components shall be filled to operating level with lubricant conforming to requirements of the applicable drawing or lubrication chart for the vehicle.

5.1.2.4 Fuel system. The entire fuel system shall be drained.

5.1.2.4.1 Storage requirement for controlled humidity (CH) storage. When vehicle is placed in CH warehouses for storage the fuel tank shall be prepared as specified in 5.1.1.7.5 with the following exceptions:

(a) Drain plug shall not be reinstalled.

(b) Filler cap shall not be reinstalled and shall be affixed to filler pipe.

(c) Drain openings and filler pipe opening shall be screened to prevent entry of insects, rodents, and miscellaneous materials and to provide ventilation of air through the fuel tank.

(d) Drain plug shall be secured adjacent to drain plug opening or secured with filler cap to the filler pipe with wire ties.

5.1.2.5 Forks. Forks shall remain in operational position.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A and B. The truck shall be shipped in its transport position unboxes (mobile). Wheels shall be free to rotate. Intermediate containers shall be positioned and secured in the truck with flat steel strapping conforming to QQ-S-781, class B, size as applicable, in a manner not to increase cubage and in a manner not to interfere with lifting the unit with

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slings or towing the unit. A tag shall be attached to the steering wheel indicating: "Towing shall be accomplished by use of a rigid towbar - Do not attempt to turn steering wheel without engine operating." The tag shall conform to UU-T-81, type A. The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.2 Level C. The truck, components, repair parts, tools, and publications shall be packed in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification Rules or with National Motor Freight Classification Rules. Towing shall be accomplished as specified in 5.2.1 and the steering wheel shall be tagged as specified in 5.2.1.

5.3 Marking. Marking of packages, intermediate containers and mobile trucks shall be in accordance with MIL-STD-129.

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured in a conspicuous location within the operator's compartment. This guide shall also require complete relubrication at all oil can points and at all grease fittings per applicable manual.

6. NOTES

6.1 Intended use. The truck described herein is intended for loading and unloading flat-cars, flat-bedded trailer, cargo aircraft, and landing craft; in stacking, unstacking, and transporting heavy crated boxes, containers, and palletized loads of heavy equipment and supplies over unprepared, unstabilized surfaces especially over rough terrain, such as over the beach, in deep sand, snow, ice and mud, where hard standing is not available and where high flotation and traction characteristics are required. The vehicles are intended for operation at any temperature range extending from plus 120°F. to minus 25°F.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) When material identification is not required (see 3.2.1).

(c) Grades of oil required at time of delivery (see 3.7).

(d) Make, model number of diesel engine and list of engine accessories to be supplied as Government furnished property. Availability of the Government furnished property (see 3.10.1 and 3.16).

(e) Parts and maintenance tools required (see 3.10.1(c), and 3.17).

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(f) When a spark arresting muffler is required (see 3.10.3).

(g) When electromagnetic interference requirements are other than as specified (see 3.11).

(h) When treatment and painting and marking other than as specified are required (see 3.14.1 and 3.15.2).

(i) Registration data required (see 3.15.2).

(j) When the engine will not be Government furnished property (see 3.16.1).

(k) When the fire extinguisher will not be Government furnished property (see 3.16.2).

(l) When rifle case is required (see 3.16.3).

(m) When technical publications other than as specified are required (see 3.18.1).

(n) When repair parts support and provisioning shall be other than as specified (see 3.19 and 3.19.1).

(o) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 The contracting officer should arrange to have the performance test specified in 4.6.22.2 conducted at a Government facility

6.4 Welding of high strength constructional steels. When high strength constructional steels are used in the fabrication of the truck, the contracting officer should require the supplier to include in the technical publications specified (see 3.2.1) the identification and location of these steels, and the ncessary precautions to be taken by the using agency in the repair and replacement of these steels. These precautions will include but not be limited to:

(a) How to avoid excessive restraint in a weld joint that could produce weld-metal or heat-affected-zone cracking.

(b) Type of weld filler metal to be used for repair of each different type of steel by the shielded Electric-Arc (coated electrode) process and by the Metal Inert Gas (MIG) process including:

(1) Type and class of electrode and MIG wire.

(2) Size (diameter) of electrode and MIG wire.

(3) Shielding medium for MIG process.

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(c) Material handling safeguards for coated electrodes and MIG wire to prevent contamination and moisture absorption, including the treatment of coated electrodes and MIG wire after exposure to contamination and moisture absorption.

(d) Preparation of base metal for welding. Method to be used (grinding or other). Preparation of root for welding second side when thickness of material necessitates that joint be welded from both sides.

(e) Base metal temperature:

(1) Minimum temperature of base metal at start of welding.

(2) Minimum and maximum preheat of base metal at start of welding.

(3) Maximum interpass temperature during welding.

(f) Rate of flow of shielding medium for MIG process.

(g) Maximum heat input during welding.

(h) The minimum ambient temperature at which welding should be done.

(i) Post weld heat-treatment (stress-relieving) requirements (necessary when weldment must retain dimensional stability during machining or stress-corrosion is a factor).

6.5 Astericks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME

Navy - SA

Air Force - 84

Preparing Activity:

Navy - SA

Project No. 3930-N225

Interested Activities:

Users and Review - Army - ME
Navy - SA - MC - OS - YD
Air Force - 84

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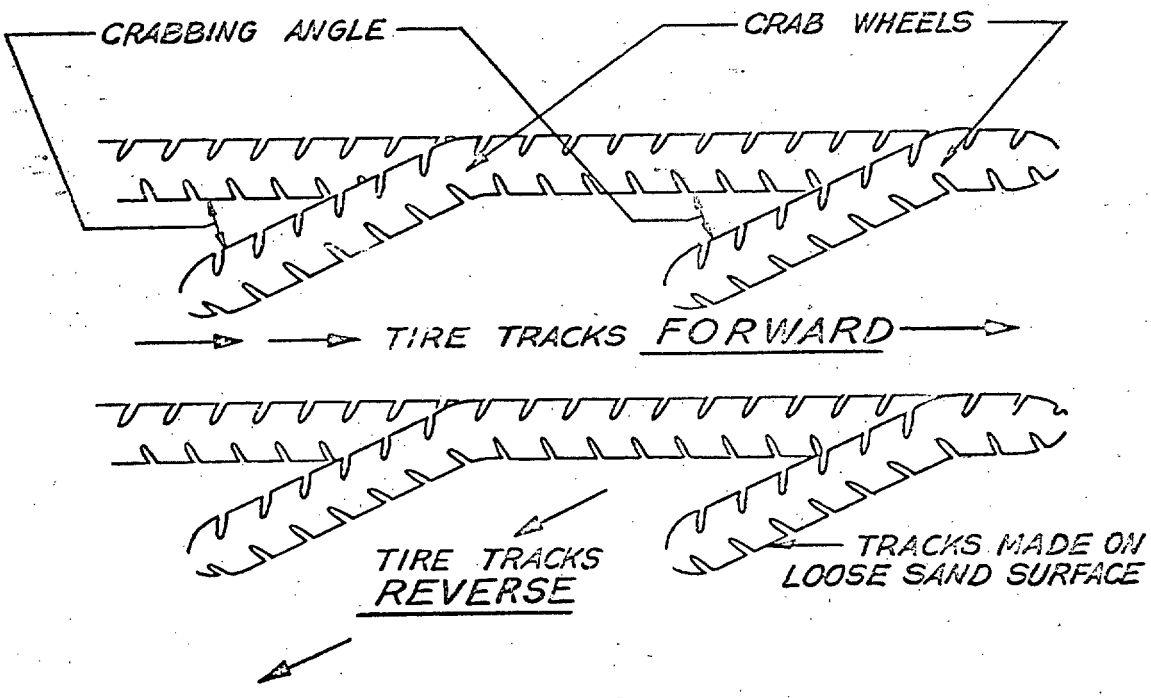


FIGURE 1

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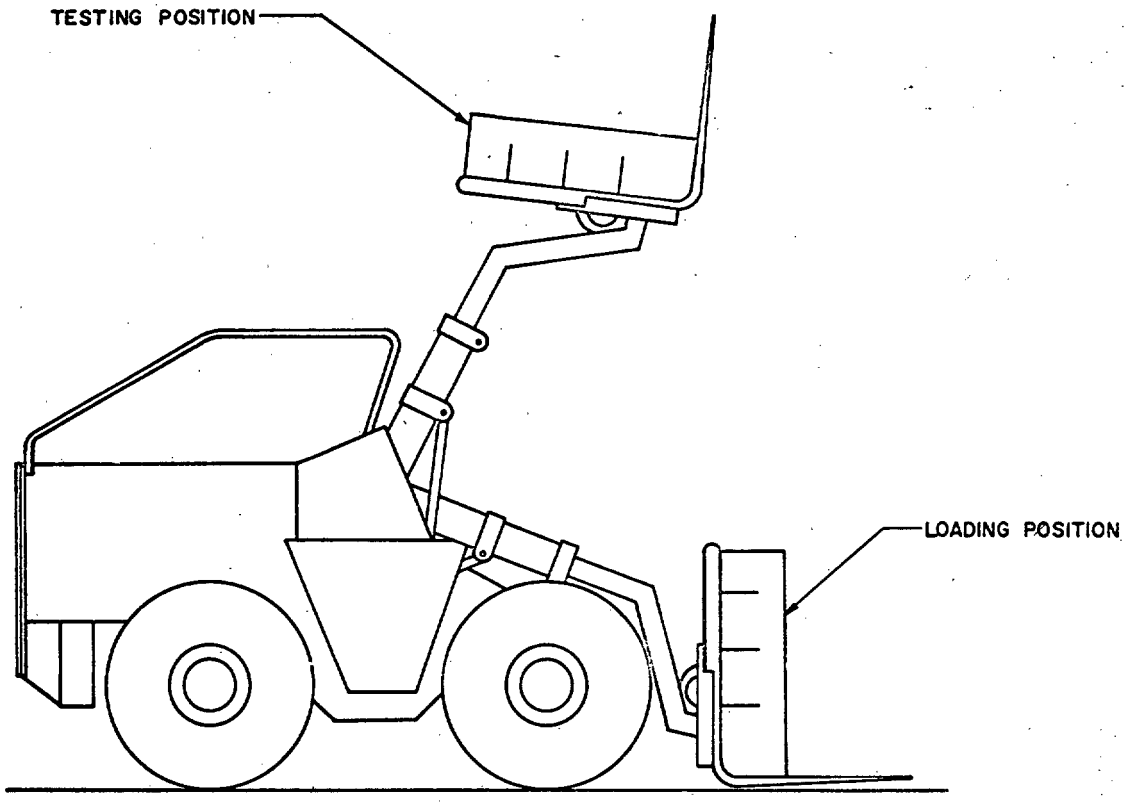


FIGURE 2 FORK RACK TEST

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APPENDIX

HYDRAULIC SYSTEM TEST

TEST METHOD NO. 1

1. TEST APPARATUS:

   a. Pressure gages.
   b. Temperature gages.
   c. Timing device accurate to 0.10 second.
   d. Tape, measuring.

2. TEST PROCEDURE:

   a. Verify each circuit pressure against specified pressure to insure proper setting.

   b. Connect each line to its related component. Operate each function of the system for 10 cycles continuously through full stroke and at maximum rates. Visually examine the filter for contamination.

   c. If there is any evidence of contamination, repeat 2a until there is no evidence of contamination. If evidence of repeated contamination exists, disassemble the truck hydraulic system, clean, and examine part by part. Reassemble and repeat the procedure specified in 2b and 2c. Reusable filters may be used for all but the last contamination check.

   d. With the system oil temperature at a minimum of 150°F., raise a rated load to a height of 6 feet at a maximum extension. Record the cylinder drift during a 15-minute period.

   e. Maintain the system oil temperature between 140°F. and 160°F., position each control lever, and pressurize each pressure line for 15 seconds. Record each relief pressure and examine for external leakage. Leakage due to defective seals and loose bolts may be corrected by replacement of the seals and tightening of the bolts to the torque recommended by the supplier providing no machining or welding is required. After repair of the leak has been made, the leakage test shall be repeated. Measure rates of lift, sideshift, and oscillation under load.

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TRAVEL SPEED TEST

TEST METHOD NO. 2

1. TEST COURSE:

The test course shall be a level, improved surface of sufficient length to attain maximum speed prior to entering measured distance of not less than 500 feet.

2. TEST APPARATUS:

a. Tape, measuring, surveyors; or odometer.
b. Watch, stop.
c. Tachometer.

3. TEST PROCEDURE:

a. Operate the truck at not more than full-load governed speed. (Note: Do not allow engine speed to increase into governor over-run.)

b. Operate the truck a sufficient distance to attain maximum speed prior to entering the measured test course.

c. Truck speed shall be the average of six runs, three in each direction.

d. When rated load is required, forks will be in carry position.

e. Conduct the following test:

(1) Speed high gear forward, empty.

(2) Speed high gear forward, with rated load.

(3) Speed high gear reverse, with rated load.

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SLOPE OPERATION TEST

TEST METHOD NO. 3

1. TEST COURSE:

   a. A 30-percent side slope of sufficient width and length to allow the truck to accelerate to a speed of 2 mph while traversing the slope.

   b. A 45-percent longitudinal slope not less than three times the wheel-base of the truck.

2. TEST APPARATUS:

   a. Inclinometer.
   b. Tape, measuring (100-foot).
   c. Watch, stop.

3. TEST PROCEDURE:

   a. Place the truck transversely on the 30-percent side slope. Start the truck without load from a standing position with the left side of the truck downhill and accelerate to not less than 2 mph while traversing the slope. Truck oscillation may be in any position advantageous to the truck. When a speed of 2 mph is attained, turn the truck full right up the slope while continuing to accelerate. Safety line may be used.

   b. Repeat the procedure in a. above with the right side of the truck downhill turning full left.

   c. Place the truck transversely on the 30-percent side slope, with rated load at 24 inch load center. Extend forks to maximum reach (zero lateral displacement). Truck oscillation shall be in the mid or normal to ground position. Check ground contact of wheels.

   d. Start the truck, without load, from a standing position at the bottom of the slope and negotiate the 45-percent longitudinal slope. Make a total of three passes up and three passes backing down the slope. On the third pass, both up and down the slope, stop the truck in the middle of the slope by applying the service brakes and releasing the accelerator. Actuate the brake holding device. Release the service brake. Wait 2 minutes. Release the brake holding device. Restart the truck by depressing the accelerator and releasing the brakes and continue to the top or bottom of the slope as applicable.

   e. Repeat the test in d. above with rated load in carry position on the fork assembly.

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LOW-TEMPERATURE TEST (MINUS 25°F.)

TEST METHOD NO. 4

1. TEST APPARATUS:

   a. Cold chamber.
   b. Arctic clothing.
   c. Thermocouples.
   d. Battery hydrometer.
   e. Wattmeter or ammeter.
   f. Recording instruments (temperature reading, electrical).
   g. Voltmeter

2. TEST PROCEDURE:

   a. Check the specific gravity of each battery to insure full charge.

   b. Subject the truck to an ambient temperature of minus 25°F. or lower until all parts, lube oil, coolant, and hydraulic oil have stabilized at minus 25°F. or lower. Perform the test after stabilization is assured by temperature readings of minus 25°F. or lower from thermocouples located as follows:

      (1) In the two center cells of each battery between the plates and at mid-depth of the electrolyte.

      (2) In the volumetric center of the hydraulic-system oil reservoir.

      (3) In the engine lubricating oil gallery and in the oil sump.

      (4) In the engine coolant in the engine-cooling-jacket discharge pipe.

      (5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the truck and not less than 3 feet from each corner of the truck at engine-crankshaft level.

   c. Maintain minus 25°F. or lower ambient temperature throughout the entire test.

   d. Start the diesel engine. Engine should run on diesel fuel in not more than 5 minutes after beginning of first cranking.

   e. One operator wearing arctic clothing including mittens shall perform all priming and starting operations.

   f. Crank the engine using batteries that were cold soaked and stabilized at minus 25°F. or lower.

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g. Substitute or supplementary batteries or other electrical cranking energy shall not be used.

h. Glow plugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the truck and normally used in starting the engine may be used.

i. Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained within 15 minutes after starting.

j. When started, run the engine 15 minutes. After completion of the 15-minute operation, operate the engine as follows:

(1) 10 minutes idling.

(2) 10 minutes half throttle.

(3) 5 minutes full throttle.

Perform the 25-minute cycle three times.

k. Record temperatures prior to engine starting, immediately after engine starting, and at 10-minute intervals during enginewarmup and cycling, including but not limited to the following:

(1) Battery.

(2) Hydraulic tank.

(3) Engine oil.

(4) Engine coolant.

(5) Chamber air.

l. During engine cycling and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels in all gears.

(2) Operate through two complete movement cycles all brakes, shafts, and cylinders.

(3) Mount to and dismount from the operator's position on the truck.

Upon completion of the cycling period, shut down the engine and (1) while truck is still in the chamber at minus 25°F. or lower, and (2) while the operator is wearing arctic mittens, restart the engine by using outside power through the slave receptacle.

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HIGH-TEMPERATURE (PLUS 120°F.) OPERATION AND ENGINE TEST

TEST METHOD NO. 5

1. TEST CONDITIONS:

The truck shall be subjected to an ambient temperature of not less than plus 120°F. until the engine coolant, fuel in the fuel tank engine lubricating oil gallery, torque converter fluid, and batteries have stablized at a temperature of not less than plus 120° F.

2. TEST APPARATUS:

a. Test chamber.
b. Thermocouples.
c. Recording potentiometer(s).
d. Thermometers (wet and dry).
e. Dynamometer with all necessary instrumentation.
f. Tachometer.
g. Barometer.

3. TEST PROCEDURE:

a. Install thermocouples as listed below:

(1) Engine coolant temperature: One in the engine-cooling-jacket discharge pipe and another in the coolant inlet to the engine.

(2) Fuel temperature: In the tank.

(3) Battery temperature: In the two center cells of each battery between the plates and at mid-depth of the electrolyte.

(4) Ambient temperature measured at an angle of 45 degrees and not less than 3 feet from each corner of the truck at engine-crankshaft level.

(5) Torque converter discharge line (to cooler).

(6) Torque converter sump.

(7) Engine oil gallery.

b. Start the engine and run for 10 minutes, then record at 10-minute inverrvals temperature readings of the thermocouples listed in 3.a. during the operations specified in 3.c.

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c. Apply for 1 hour a load equal to 90 percent of the intermittent brake horsepower at governed speed as determined by the engine supplier. At the end of the 1-hour period, change the load and cycle as follows for 10 complete cycles:

(1) Maximum rated load (as determined above) - 3 minutes.

(2) 1/2 maximum rated load - 2 minutes.

(3) Zero load - 1 minute.

d. Record the following readings at 10-minute intervals during the 2-hour operation:

(1) Revolutions per minute.

(2) Dynamometer torque.

Take readings (1) and (2) simultaneously.

(3) Barometer.

(4) Wet-and Dry-bulb air temperatures.

e. Operate all controls and switches other than starting switch.

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LIFTING-ATTACHMENT TEST

TEST METHOD NO. 6

1. TEST APPARATUS:

   a. Crane or lifting device.
   b. Slings with hooks.

2. TEST PROCEDURE:

   a. Attach the slings to the truck, lifting eyes in such a manner that they converge at a point not more than 19 feet above the lowest extremity of the truck. Lift the truck and hold in suspension in its normal traveling position. Spreader bars are permitted. Determine the force and the angles of application for each lifting attachment. Measure the clearance between each sling and the truck.

   b. Restrain the truck by anchoring the main frame or axles and subject each lifting attachment to a load applied in the direction as determined in 2.a. and equal to 2-1/2 times the force determined in 2.a. Hold each load for a minimum of 90 seconds.

   c. An alternate method of performing the above, is to secure a load sled with such additional weight secured to the seld as necessary to total 1-1/2 times the shipping weight of the truck. The sled and load shall be placed so that the truck and sled when lifted with slings as in 2.a. above will be level or normal to the ground. Hold the truck with load for a minimum of 90 seconds.

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STATIC LOAD TEST

TEST METHOD NO. 7

1. TEST COURSE:

The test course shall be a level surface.

2. TEST APPARATUS:

a. Load equal to rated load.
b. Load equal to twice rated load.
c. Load equal to three times rated load.

3. TEST PROCEDURE:

a. Place the truck on the test course and anchor the rear of the truck to the ground. The relief and replenishing valve may be taken out. Position the forks so that they are 30 inches apart (outside to outside), laterally centered on the carriage, raised 30 inches above ground level, and extended 21 inches in front of the tires. Place three times rated load on the forks and leave for a minimum period of 10 minutes. Remove the load and examine the truck.

b. With the truck on the test course and the forks positioned as in 3.a. except laterally displaced to the extreme right, place twice rated load on the forks. Leave the load on the forks for a minimum of 5 minutes. Remove the load and laterally displace the forks to the extreme left position. Place the load on the forks and leave for a minimum of 5 minutes. Remove the load and examine the truck.

c. Repeat the test in 3.b. except place rated load on each of the forks. Remove the load and examine the truck.

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ENDURANCE (DURABILITY COURSE)

TEST METHOD NO. 8

1. TEST COURSE:

A test course shall be constructed in accordance with fig A-1 The total length of the course shall be 750 feet, plus or minus 30 feet.

2. TEST PROCEDURE:

Drive the truck, with rated payload in carry position, for 300 hours over the test course. Speed over the crosstie and pothole portions of the course shall be not less than 3.5 mph. Negotiate the course 50 percent of the time in a clockwise direction and 50 percent of the time in a counterclockwise direction. Negotiate the course at least 25 times per hour. During the second 100-hour period, travel over potholes shall be in 4-wheel drive and travel on flat stabilized surface shall be in 2-wheel drive. During the last 100 hours of the test, all turns shall be in 4-wheel steer and all travel over crossties and pothole areas shall be in 2-wheel steer. After each 8 hours, operate the side shift at full speed and full load for 10 full-stroke cycles.

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ENDURANCE (LOAD PLACEMENT)

TEST METHOD NO. 9

1. TEST COURSE:

The test course shall be level, improved surface laid out in accordance with fig. A-2.

2. TEST APPARATUS:

a. Palletized loads (three each, rated load).
b. Platforms (two each, 8 feet high).

3. TEST PROCEDURE:

a. Position the platforms at points 3 and 2 on course "A".

b. Position the loads at points 1, 4, and 5 on course "A". Place a block under one side of loads 4 and 5 so that load 4 will be tilted approximately 6 degrees to the right and load 5 will be tilted approximately 6 degrees to the left.

c. Start at point 1, pick up the load, and move to point 3. Place the load on the platform at point 3 and proceed to point 5. Pick up the load at point 5 and move it to point 2. Place the load on the platform at point 2 and proceed to point 4. Pick up the load at point 4 and move it to point 1. Continue the cycle moving from point 1 to point 3, from point 3 to point 5, etc. picking up or placing the loads as applicable for a total of 35 hours. The truck shallmaneuver within the area of the course during the operations. A minimum of 108 tons shall be moved each hour.

d. Repeat the test in a., b., and c. on course "B".

e. Repeat the test in a., b., and c. on course "C" except that the time shall be reduced to 30 hours.

f. Each load shall be picked up at full extension and before depositing the load the inner boom shall touch the main frame.

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ENDURANCE TEST (SEA WATER)

TEST METHOD NO. 10

1. TEST COURSE:

The test course shall be a sandy beach area and a body of sea water of sufficient depth to permit submersion of the truck to a depth of not less than 5 feet.

2. TEST PROCEDURE:

a. Paint a horizontal white line on each side of the truck at a height of 5 feet above ground level with rated load on the forks.

b. Operate the truck, with rated load, as follows:

(1) Drive the truck into the water to a depth of 5 feet including wave action. Maneuver the truck, for a period of not less than 15 minutes, parallel to the beach and toward the beach to a minimum water depth equal to the centerline height of the wheels. Maneuvering shall consist of the following:

(a) Drive the truck forward in first gear at maximum safe speed for a period of 3 minutes. Stop the truck by applying the service brakes.

(b) Drive the truck backward in low reverse at maximum safe speed for a period of 3 minutes. Stop the truck by applying the service brakes.

(c) During the 6-minute cycle in (a) and (b) above, the steering system shall be actuated from stop to stop in each of the three modes of steer.

(d) At the end of the 6-minute cycle and while at maximum water depth, operate all hydraulic cylinders, except steering cylinders, through one complete movement cycle.

(2) At the end of the 15-minute period, drive the truck onto the beach and repeat the maneuvering at maximum attainable speed for a period of 15 minutes.

c. Repeat the procedure in 2.b. for a total of 8 hours per day for 2 days. Prior to leaving the water for the last time during the 8-hour period and while at maximum water depth, shut-down the engine and restart it.

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d. At the end of the first 8-hour period and after scheduled servicing, the truck shall be parked on the beach in an unprotected location and left until start of the next 8-hour period.

e. Park the truck in an unprotected location on the beach for 10 days.

f. Repeat the test in 2.c. for one additional 8-hour day.

g. Oil samples shall be taken from the lowest point of the following at the beginning of the test, at the completion of one day of test, at the completion of 2 days of test, at the completion of 12 days of test, and at the conclusion of the test:

(1) Torque converter sump.

(2) Engine oil sump.

(3) Transmission

(4) Fuel tank.

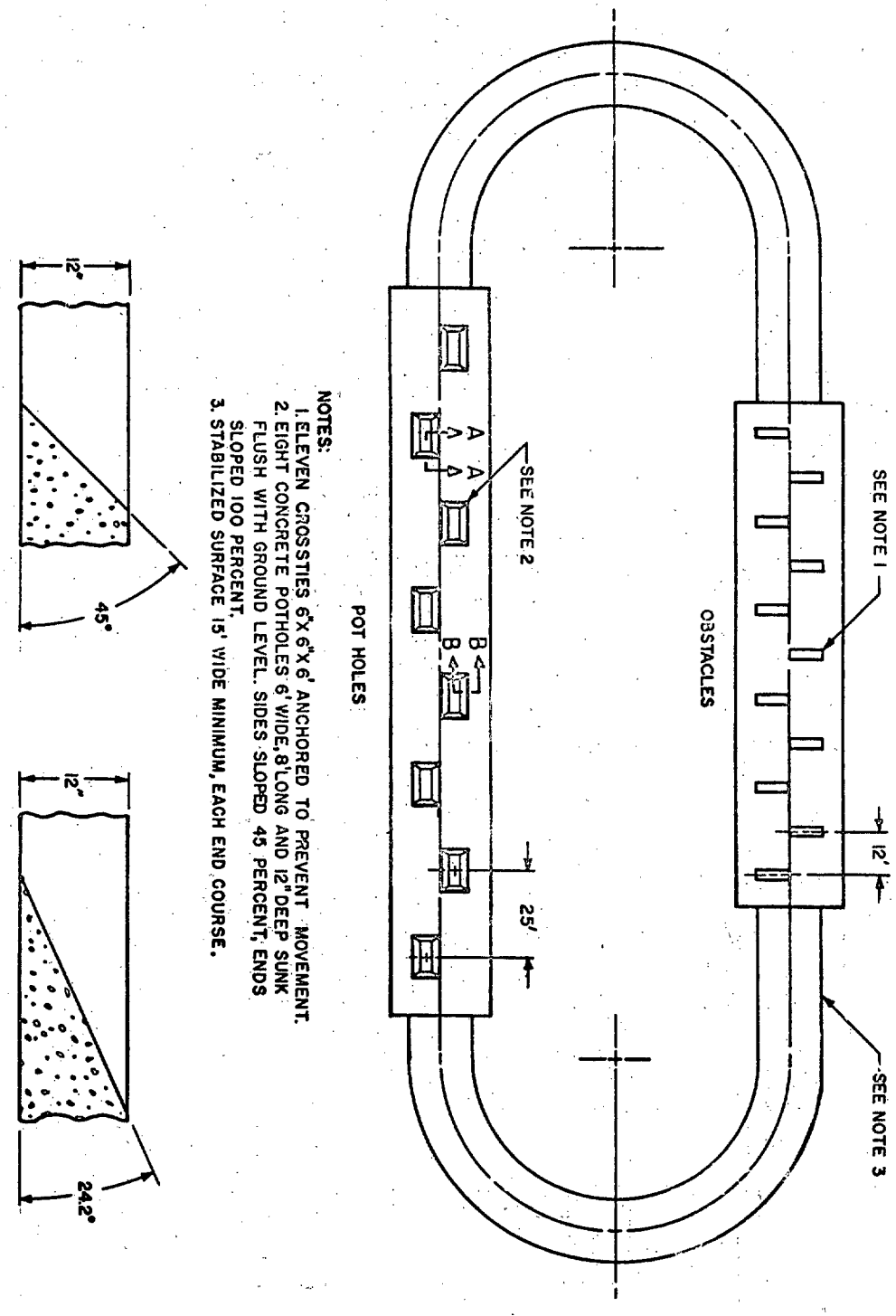
(5) Hydraulic reservoir.

(6) Each axle.

(7) Each wheel hub.

Samples, except for those taken at the beginning of the test and at the completion of 12 days of test, shall be taken 1 hour after the truck has been shut down and prior to any servicing. Samples shall be analyzed for water and salt contamination.

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NOTES:
1. ELEVEN CROSSTIES 6"X 6"X 6' ANCHORED TO PREVENT MOVEMENT.
2. EIGHT CONCRETE POTHOLES 6' WIDE, 8' LONG AND 12" DEEP SUNK FLUSH WITH GROUND LEVEL. SIDES SLOPED 45 PERCENT, ENDS SLOPED 100 PERCENT.
3. STABILIZED SURFACE 15' WIDE MINIMUM, EACH END COURSE.

FIGURE A-1. DURABILITY COURSE

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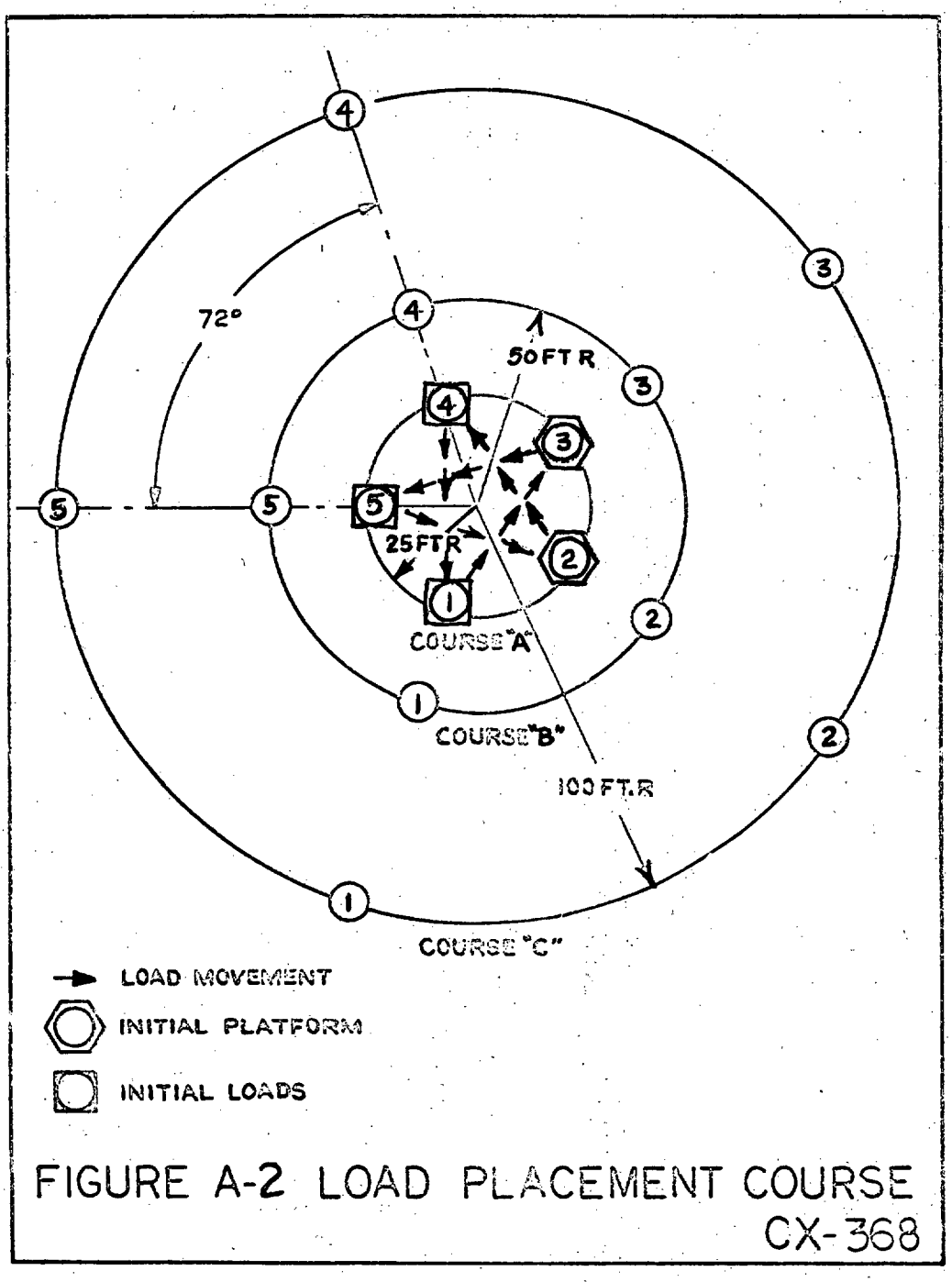


FIGURE A-2 LOAD PLACEMENT COURSE
CX-368

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